



A branch-and-price heuristic algorithm for the ART and external truck scheduling problem in an automated container terminal with a parallel layout

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ARTICLE INFO

Keywords:

Automated container terminal
Horizontal transportation equipment
scheduling
Branch-and-price
Column generation

ABSTRACT

The scheduling of horizontal transportation equipment in automated container terminals is a focal area of concern in the port industry. In an automated container terminal with a parallel layout, artificial intelligence robots of transportation (ARTs) and external trucks (ETs) are critical vehicles that connect the quayside, yard side, and hinterland. The operational management of ARTs and ETs is complex because of various factors affecting their operational performance, such as vehicle congestion, limited handover point capacity, limited crane operation capacity, and intersection of ARTs and ETs. This study considers the ART and ET scheduling problem, which involves the task assignment of ARTs and the sequencing of all operations for ARTs and ETs. We formulate the problem as a mixed-integer linear programming (MILP) model to minimize the delay time for all ART and ET tasks. However, since the problem is NP-hard, the MILP model cannot be efficiently solved for realistic-scale instances. Therefore, we reformulate the original model into a route-based model and propose a tailored branch-and-price heuristic algorithm to solve the new formulation. In addition, several acceleration methods adapted to the problem characteristics are introduced to enhance the performance of the algorithm. Computational experiments validate the effectiveness of the proposed algorithm and provide managerial insights to support operational decision-making for terminal operators.

1. Introduction

Container terminals, as pivotal points in global trade and transportation, play a crucial role in facilitating the efficient handling and transfer of goods between various modes of transport. With the increasing adoption of intelligent and information technologies to enhance operational efficiency and service levels of container terminals, the construction of automated container terminals (ACTs) has become an inexorable trend (Li et al., 2023a). Over the past decade, the number of ACTs worldwide has experienced exponential growth (Naeem et al., 2023), accompanied by increasing diversification of ACT layouts. The classic layout features yard blocks perpendicular to the shoreline, whereas the layout of yard blocks parallel to the shoreline has been widely applied in recent years, such as the newly constructed ACTs in Tianjin Port and Guangzhou Port, China.

Perpendicular layouts typically include the classic perpendicular layout (Naeem et al., 2023) and the U-shaped perpendicular layout (Niu et al., 2022). In the classic perpendicular-layout ACT, the transportation areas for internal trucks and external trucks (ETs)

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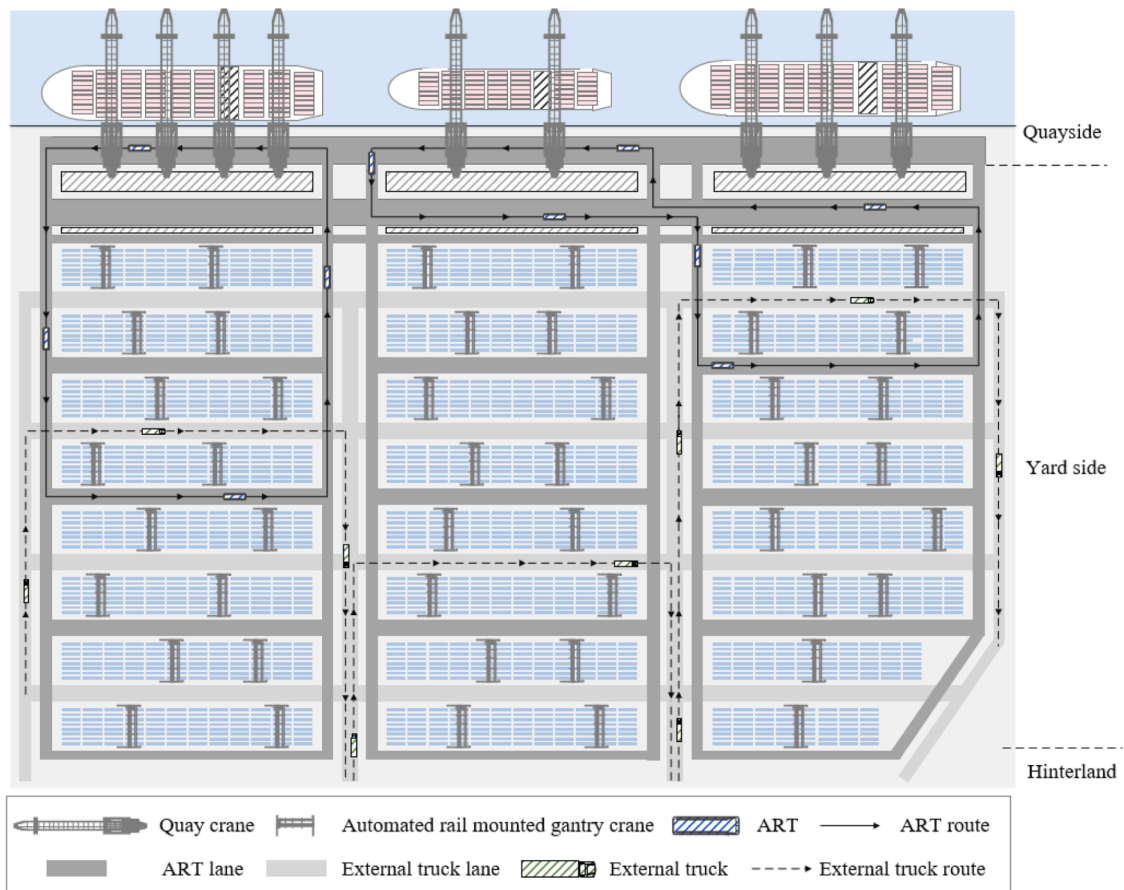


Fig. 1. A typical ACT with a parallel layout.

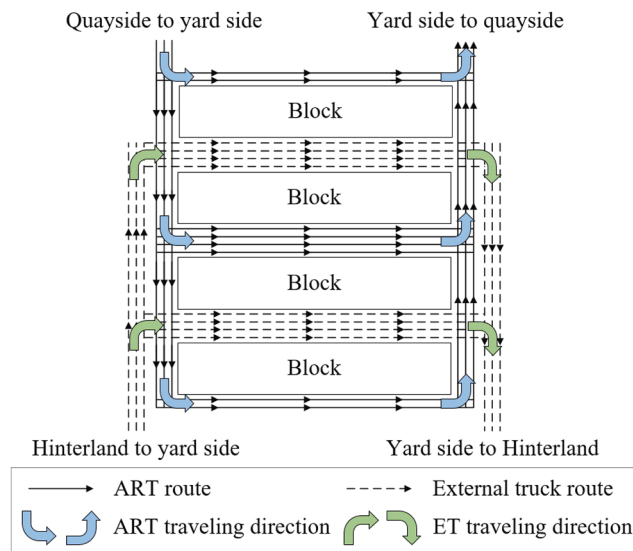


Fig. 2. The transportation areas of ARTs and ETs in the yard.

are located at opposite yard ends, such as the Qingdao Qianwan ACT in China. In the U-shaped ACT, although internal trucks and ETs can access the yard block from the side, they travel on completely different lanes, such as the Guangxi Qinzhou ACT in China. Therefore, in perpendicular-layout ACTs, the transportation areas of internal trucks and ETs are physically separated, representing two

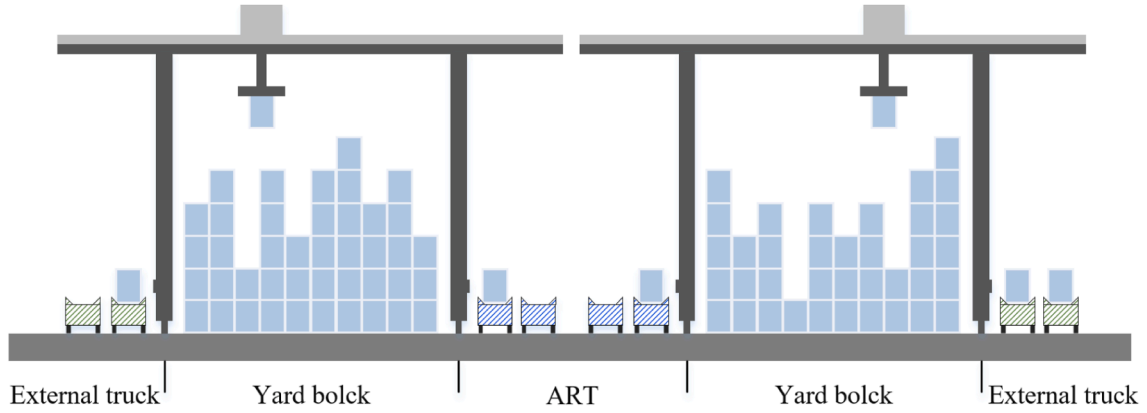


Fig. 3. Isolated lane layout of the yard.

relatively independent processes. However, in parallel-layout ACTs, the travel of internal trucks and ETs is an integrated process. Specifically, the parallel-layout ACT has lanes on both sides of the yard block, where internal trucks and ETs travel on separate lanes. In this layout, the interaction between internal and external trucks occurs in the vertical lanes (see Fig. 1 to Fig. 3 for details). In addition, it should be pointed out that conventional parallel-layout container terminals typically have lanes on only one side of the yard block (Lee et al., 2018), and the interaction occurs across all lanes.

A typical ACT with a parallel layout, the Beijiagang Section C automated container terminal in Tianjin Port, China, is shown in Fig. 1. Quay cranes (QCs) and automated rail-mounted gantry cranes (ARMGs) are responsible for container exchange operations on the quayside and yard side, respectively. Artificial intelligence robots of transportation (ARTs) and ETs are two types of vehicles operating within the terminal. ARTs handle container transportation between the quayside and yard side, whereas ETs serve as the key link between the yard and hinterland. The travel lanes of both the ARTs and ETs are unidirectional, as shown in Fig. 1 and Fig. 2. As presented in Fig. 2, ARTs and ETs travel on separate lanes. However, intersections occur where the ART and ET lanes meet. ART and ET lanes, with the capability of accommodating two ARTs and two ETs side-by-side, are present on either side of the yard block, as depicted in Fig. 3. Additionally, the positions at which the QCs and ARMGs perform container exchanges with vehicles are called handover points (HPs), which have a limited number of vehicles that can be accommodated simultaneously. Effective scheduling of ARTs and ETs is crucial for ensuring efficient crane operations and ultimately improving terminal productivity. However, vehicle movement in the ACT is extremely complex, as it involves not only the interaction between ARTs and ETs but also congestion within their respective lanes. Therefore, the critical challenge faced by ACT is scheduling ARTs and ETs efficiently with limited lane resources, thus accomplishing effective coordination with cranes.

Optimizing vehicle transport operations to ensure efficient and timely service of cranes is considered a cost-effective solution for coping with the rapid growth of container throughput (Li et al., 2023a). The first decision for vehicle operation optimization is the ART scheduling, which is essential for facilitating efficient operations on both the quayside and yard side. The number of ARTs that can be accommodated by resources, such as cranes, lanes, and HPs, is limited, and an excessive number of ARTs simultaneously occupying the same resource can lead to resource occupancy conflicts. Thus, ART scheduling requires decisions not only on the task assignment and operation sequence but also on the timing of container exchange and resource occupation, considering the limited resource capacity.

The second decision of vehicle operations involves ET scheduling. ET scheduling primarily determines the timing for ETs to perform container exchanges with ARMGs and occupy ET lanes and HPs. ARMGs serve both ETs and ARTs, which complicates the problem because ART and ET operations are mutually interdependent, considering that an ARMG cannot perform container exchange with an ART and an ET simultaneously. Thus, the scheduling decisions for ETs affect the timing for ARMGs to exchange containers, whereas ARMG operations affect the ART scheduling scheme, which in turn affects the ET decisions. Additionally, the intersection of the ART and ET lanes (see Fig. 1 and Fig. 2) further emphasizes the close relationship between ART and ET scheduling.

This study addresses the ART and ET scheduling problem (AESP) in an ACT with a parallel layout, which is especially significant nowadays as the development of such ACTs has attracted widespread attention over the years. The main contributions of this study are as follows. First, this study proposes a vehicle scheduling problem at the operational level in a real ACT with a parallel layout. Considering vehicle congestion, limited HP capacity, limited crane operation capacity, and the intersection of ART and ET lanes, we address the operation management problem of ARTs and ETs. Second, we present a mixed-integer linear programming (MILP) model for the AESP. In this model, the resource occupancy conflict, which ensures that the number of vehicles occupying each resource (path, HP, or crane) does not exceed its capacity in each time step, is proposed to guarantee the HP capacity and crane operation capacity and to avoid vehicle congestion. Furthermore, the model captures the effects of the ART and ET on each other, including their simultaneous service at the ARMG and the intersection between the two types of vehicle lanes. Third, we reformulate the MILP model as a route-based formulation based on the analysis of the model structure and feature. To solve this formulation, a tailored branch-and-price heuristic (BPH) algorithm is developed, in which ART and ET are portrayed as two separate pricing sub-problems. Several acceleration methods are devised to improve the performance of the algorithm by exploiting problem characteristics. Moreover, we conduct computational experiments to compare the performance of the proposed algorithm with that of existing methods and investigate the

effectiveness of the acceleration methods. Sensitivity analysis experiments are conducted to evaluate the influence of key parameters on the results, yielding valuable managerial insights. Specifically, a real-world case study is conducted to validate the applicability of the BPH algorithm to dynamic problems.

The remainder of this paper is organized as follows. Section 2 provides a review of the relevant literature. Section 3 describes the problem and presents the MILP model along with an explanation of the problem complexity. Section 4 introduces route-based formulation and describes the development of the BPH algorithm. Computational experiments are presented in Section 5 to validate the effectiveness of the BPH algorithm and the practicality of the proposed approach. Section 6 discusses managerial insights. Finally, Section 7 summarizes the study and outlines future research directions.

2. Literature review

This study investigates the operational management and optimization of ARTs and ETs in an ACT with a parallel layout, which is closely related to the extensively studied problem of equipment scheduling in ACTs. The ACTs with different layouts have different equipment types and operational modes. To assist in layout planning and facility selection for ACTs, Li et al., (2023b) proposed a criteria system and introduced an agent-based simulation method in which a multi-criteria decision-making framework and a new improved technique were designed to evaluate ACTs with different layouts. For ACTs with different layouts and facility types, some studies have developed customized models and algorithms that adapt to problem characteristics. Several studies have focused on equipment scheduling problems in ACT with a perpendicular layout, where vehicles operate at the ends of yard blocks, such as quay crane scheduling, automated guided vehicle (AGV) scheduling, yard crane scheduling, and integrated scheduling (Jenny et al., 2018; Chen et al., 2020; Li et al., 2020; Zhuang et al., 2022). For the U-shaped layout, where the yard is perpendicular to the shoreline and vehicles operate on the sides of the yard blocks, some studies have developed tailored heuristic algorithms to solve the multi-equipment coordinated scheduling problem (Xu et al., 2021a; Niu et al., 2022; Liu et al., 2023b). A few studies have addressed the equipment-scheduling problem in ACT with a parallel layout, where vehicles operate on the sides of yard blocks (Tang et al., 2021; Xu et al., 2021b). Wu et al. (2022) introduced a vehicle-scheduling framework model based on the Petri network and A-star algorithm, in which a parallel layout is one of the operational modes. Additionally, several other layout forms of ACTs, such as the frame-bridge-based ACT and GRID-based ACT, have garnered attention (Jiang et al., 2018; Zhen et al., 2018; Zhou et al., 2018).

The internal truck scheduling problem is a major area of research on ACTs (Carlo et al., 2014). AGVs have received extensive attention as some of the most widely used internal trucks in ACTs. Some studies have focused on the task assignment of AGVs without considering conflicts and congestion (Luo and Wu, 2015; Choe et al., 2016; Hu et al., 2019; Zhuang et al., 2022; Li et al., 2023a), which is critical for AGV scheduling in practice. Numerous methods have emerged in recent years to tackle AGV conflicts and congestion, including the branch-and-bound heuristic algorithm, ant colony algorithm based on multiagent reinforcement learning, and other heuristic algorithms (Adamo et al., 2018; Hu et al., 2023; Wang and Zeng, 2022; Wu et al., 2022; Xu et al., 2021a; Zhong et al., 2020). Yue and Fan (2022) proposed a dynamic scheduling method to address AGV scheduling and path planning problems. However, most of the studies that consider conflict and congestion focus on ACTs with perpendicular layouts, and only a few studies have focused on vehicle path planning in ACTs with parallel layouts, which is addressed in this study.

Another stream of internal truck scheduling in ACTs involves the integrated scheduling of internal trucks with other decision components. To solve the joint scheduling problem of QCs and AGVs, Yue et al. (2021) introduced a two-stage multi-objective model to minimize the QC delay time and AGV ineffective time while maximizing customer satisfaction. Considering the limited buffers, Zhuang et al. (2022) tackled the joint scheduling problem of intelligent handling equipment under bidirectional flows and developed an adaptive large neighborhood search algorithm. Furthermore, some studies have addressed the integrated scheduling problem of vehicle dispatching and storage allocation (Luo and Wu, 2015; Hu et al., 2019), integrated scheduling problem of all container handling equipment (Roy and De, 2018; Yang et al., 2018; Li et al., 2020; Luo and Wu, 2020), and operation management problem considering power management and battery swapping (Xiang and Liu, 2021; Li et al., 2023a). The existing literature on integrated scheduling does not consider the connection between the internal and external trucks, whereas this factor is considered in this study.

The turnaround time of ETs is an important indicator for evaluating the performance of a container terminal and is closely related to their arrival and departure times (Abdelmagid et al., 2022). Existing studies on the ET appointment-scheduling problem have mainly focused on three main classes: control and decision perspectives, modeling methodologies, and collaboration among different terminals (Li et al., 2018; Caballini et al., 2020; Azab et al., 2020). By utilizing the partial arrival information of external trucks, Zeng et al. (2019) solved the joint scheduling problem of pickup sequences and container rehandling in container terminals. Furthermore, the operational management of ETs in ACTs has attracted academic attention. To overcome the limitation that traditional methods of ET assignment are inadequately applicable, He et al. (2023) considered the operational characteristics of ETs in ACTs with perpendicular layouts and proposed an optimization model for ET appointment and ARMG scheduling. However, most studies on ET operation management still focus on traditional terminals, and few have been adapted to ACTs with parallel layouts. Accordingly, we consider the impact of ETs on vehicle transportation management and address the internal truck (ART) and ET scheduling problems in an ACT with a parallel layout.

As mentioned above, the review of related studies can be mainly summarized as follows. First, existing studies on equipment scheduling of ACTs have predominantly focused on perpendicular layouts, while our emphasis lies on the vehicle scheduling problem within a parallel layout. The operation management of parallel-layout ACTs remains a relatively new and underexplored topic that has not been systematically investigated. Second, a few studies have simultaneously considered factors such as vehicle congestion, limited HP capacity, and crane operation capacity, which are crucial for the practical applicability of existing methods. Third, in parallel-layout ACTs, the operation management of ETs is yet to be considered in relation to the impact on internal truck operations. In

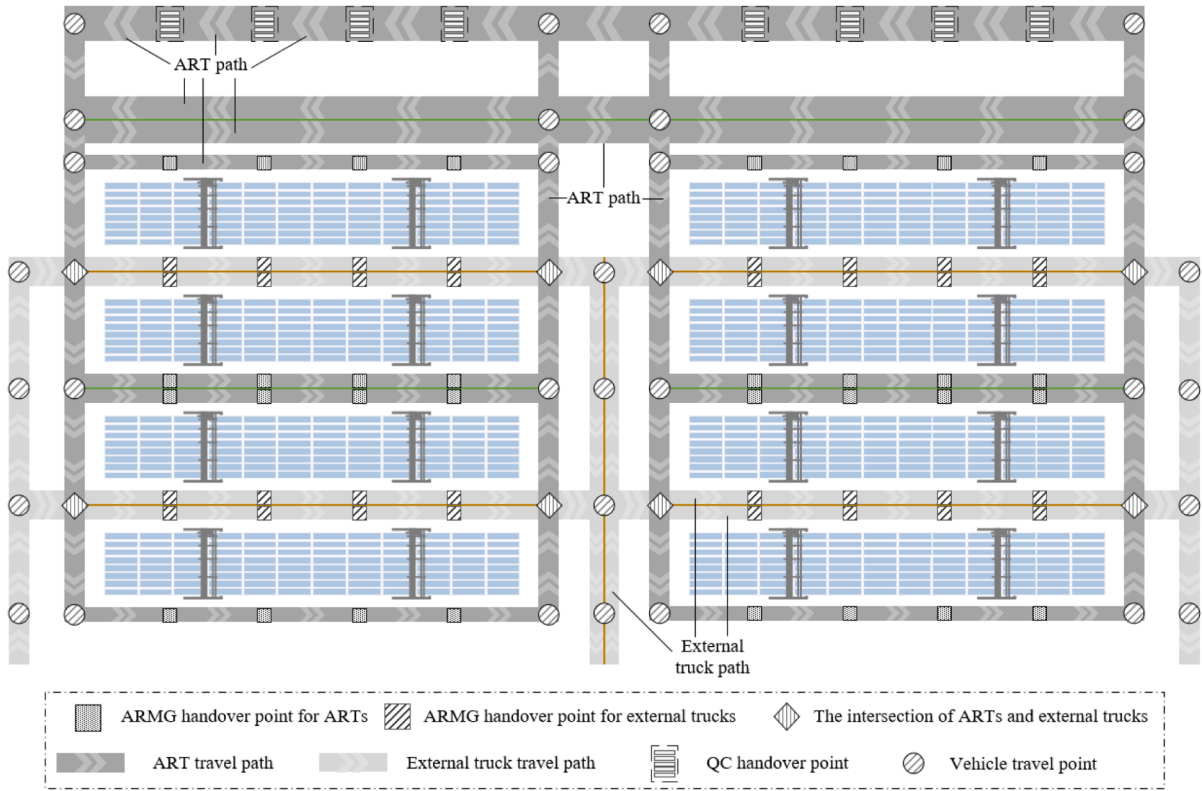


Fig. 4. The transportation area of ARTs and ETs.

particular, the existing methods used in perpendicular layouts cannot be directly applied to the vehicle scheduling in parallel layouts with minor adaptations, mainly due to the following two reasons. 1) In ACTs with perpendicular layouts, the transportation areas for internal and external trucks are physically separated. Existing methods mainly focus on the scheduling of internal trucks and ignore ETs, making it challenging to address the interaction of internal and external truck movements in parallel layouts. 2) While a considerable number of existing methods have addressed various problems related to internal trucks, there is limited research that simultaneously considers factors such as vehicle congestion, limited HP capacity, and limited crane operation capacity.

To overcome these gaps, we investigate the ART and ET scheduling problem in a parallel-layout ACT, considering the limited resource capacity and the intersection between ARTs and ETs. An MILP model and a tailored BPH algorithm are proposed to solve the problem, and several acceleration methods applicable to the problem characteristics are introduced. Computational experiments validate the effectiveness of the BPH algorithm and provide managerial insights through sensitivity analysis. It should be noted that, as demonstrated in the real-world case study in Section 5.4, our approach can be incorporated into rolling horizon planning to adapt to the dynamic environment of the ACT, where the planning can be rolled to the next stage as soon as new information or data becomes available.

3. Problem statement and formulation

3.1. Problem description

Consider an ACT with a parallel layout where ARTs and ETs collaborate while handling the container flow between the ship, yard, and hinterland, and would like to provide an efficient vehicle operation scheme to accommodate the operational demand of cranes. Typically, before the arrival of container ships and ETs, the operational plans of QCs and ARMGs are generated (Li et al., 2016; Zheng et al., 2019; Li et al., 2023a). Subsequently, based on these plans, the terminal operators need to determine the task assignment and operation sequence for ARTs as well as when ART and ET tasks are executed to minimize the delay time of all tasks. However, several practical factors render this task challenging.

- The crane operation capacity is limited because only one task can be performed by a QC or ARMG at a time. Therefore, the timing of container exchanges between vehicles and cranes cannot rely solely on the timing of vehicles reaching the crane positions.
- The movement of a large number of vehicles (ARTs and ETs) can lead to vehicle congestion because multiple vehicles inevitably occupy the same path resource simultaneously, which prevents subsequent vehicles from accessing that path and thus causes queuing.

- Each area where vehicles and cranes exchange containers can be modeled as a handover point (HP), and the number of vehicles that can be accommodated simultaneously at each HP is limited.
- ARTs are heterogeneous, and each ART has an initial position and ready time corresponding to its unfinished task, which can be characterized by a dummy beginning task.
- The ART and ET lanes have intersections, as shown in Fig. 1, which means that if both types of vehicles arrive at an intersection point simultaneously, one vehicle must wait for the other to pass first. In addition, because an ARMG can handle only one task at a time, delays are inevitable for ARTs and ETs serving the same ARMG simultaneously. Consequently, decisions regarding ART and ET operations should be made to optimize the utilization of the available resources. Otherwise, separate decisions for ARTs and ETs may result in suboptimal or infeasible solutions.

To incorporate the aforementioned realistic characteristics into the proposed model and algorithm, we first define the resources that include QCs, ARMGs, paths, and HPs. The resource capacity refers to the maximum number of vehicles that each resource can simultaneously accommodate. As shown in Fig. 4, we mark the handover points for container exchange between the QCs and ARTs, ARMGs and ARTs, and ARMGs and ETs, which are called the QC handover point, the ARMG handover point for ARTs, and the ARMG handover point for ETs, respectively. Note that for ease of modeling, we define the location of vehicles and cranes to exchange containers at the handover points. The number of handover points for each yard block is determined based on the block and vehicle lengths. The vehicle travel points and intersections of the ARTs and ETs are marked in Fig. 4, where the vehicle travel points are introduced to divide the travel lanes of the ARTs and ETs into multiple paths. In addition, the entire planning horizon is divided into a set of discrete time steps. Accordingly, we introduce the concept of “resource occupancy conflict”, which ensures that the number of vehicles occupying each resource does not exceed its capacity in each time step. The application of the resource occupancy conflict involves restricting the number of vehicles simultaneously present within a resource, ensuring the limitations of the crane operation capacity and HP capacity while avoiding vehicle congestion (the number of vehicles in each path is limited). The concept of “vehicle conflict” is also defined, which occurs if the ART and ET simultaneously arrive at the intersection of their lanes, requiring one vehicle to wait for the other to pass.

Based on the unidirectional path structure of the ART and ET lanes, as illustrated in Fig. 1 and Fig. 4, we introduce the following Proposition 1 to determine the travel routes of the vehicles. The lanes under the QCs and ARMGs are accessible only to vehicles that exchange containers with these cranes. Because the container storage locations on the ship and in the yard are given, this implies that the quayside and yard-side HPs for ART tasks, as well as the HPs for ET tasks, are determined. Therefore, the loaded travel paths for the ART tasks and all the travel paths for the ET tasks are predetermined. Moreover, the ART scheduling process can be represented on a graph (N, A) , where N is the node set of ART tasks, and A is the set of ART arcs. Arc $(i, j) \in A$ denotes a consecutive container pair that can be executed by an ARTs. For each arc (i, j) , the end HP of task i and start HP of task j are known, which means that if arc (i, j) is visited by an ART, the corresponding empty travel paths are determined. Accordingly, we use arc (i, j) to represent ART empty travel from tasks i to j . In summary, the decisions in this problem do not involve the selection of vehicle paths as ART paths are determined based on the assignment of ART tasks, and ET paths are inherently fixed.

Proposition 1. (If the start HP and end HP of an ART task are known, then the ART travel paths between these two HPs are known, and if the ARMG handover point for an ET is known, then the travel paths of the ET are known.) **Proof.** The proof is presented in Appendix A.

Each container task has a soft time window (task ready time and due time) for service, which is determined by the scheduling of the QCs and ARMGs (Lee et al., 2009). A task can only be performed after its ready time, and the due time of the task can be violated with a penalty, that is, the delay time. The ready time for the ET task is also related to the ET arrival time. Note that both unloading and loading operations are taken into consideration. For ease of modeling, we define the first and second crane operations for ART tasks. For an unloading container, the first crane operation corresponds to the container exchange between an ART and a QC (ART quayside operation), while the second crane operation is the container exchange between an ART and an ARMG (ART yard side operation). The loading containers have the first and second crane operations in the opposite manner to unloading containers. Additionally, for the unloading operations of ARTs from the seaside to the yard side, the start HP and end HP are the seaside HP under QCs and the yard-side HP under ARMGs, respectively. For loading operations, the start HP and end HP are the yard-side HP and the seaside HP, respectively.

We further define an operation set O that contains three types of operations: vehicle occupancy cranes, HPs, and paths, where $O = \{o | o \in O_i, i \in C^T\} \cup \{o | o \in O_{ij}^E, (i, j) \in A\}$. Specifically, for ART task i with n_1 loaded paths, the operation set O_i includes the first crane operation o_i^F , the second crane operation o_i^S , the ART occupying HP operation o_i^A of the first crane operation, the ART occupying HP operation o_i^B of the second crane operation, and the set O_i^L consisting of n_1 loaded travel operations corresponding to n_1 loaded paths. Similarly, for arc (i, j) with n_2 empty paths, the operation set O_{ij}^E is composed of n_2 empty travel operations corresponding to n_2 empty paths. For ET task i with n_3 travel paths, the operation set O_i includes the yard operation o_i^Y , the ET occupying HP operation o_i^C , and the set O_i^L consisting of n_3 travel operations corresponding to n_3 travel paths. The vehicle occupying HP operation and the crane operation are distinct as a vehicle may enter and occupy an HP in advance to wait for the crane operation. In set O , an operation can only be executed in a specific known resource, whereas a resource can accommodate multiple operations. Thus, the conflict of multiple vehicles occupying a single resource can be transformed into the conflict where multiple operations are simultaneously performed on a single resource. Additionally, the vehicle conflict between ARTs and ETs can be transformed into path pairs, where a path pair refers to two paths that generate an intersection.

To illustrate the travel path forms of the ART and ET tasks, we provide an example with 39 paths, as shown in Fig. 5. In this

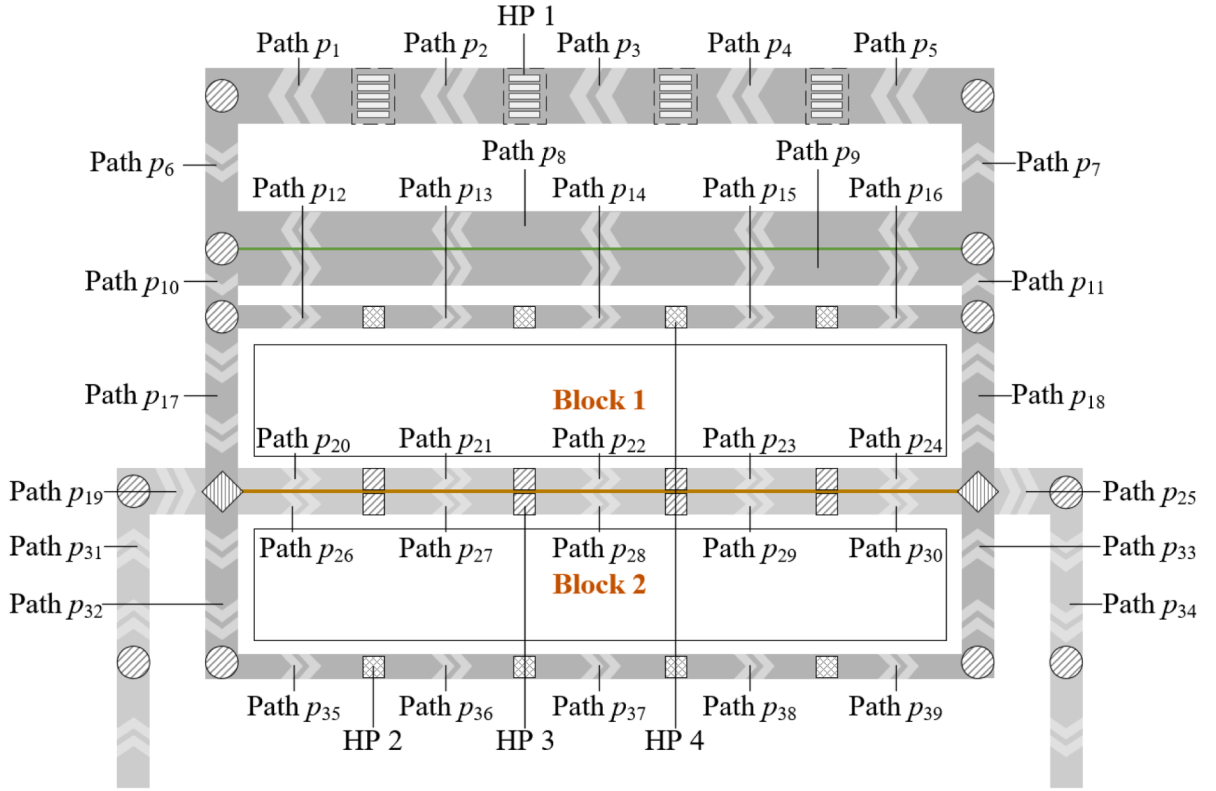


Fig. 5. An example of vehicle transportation area.

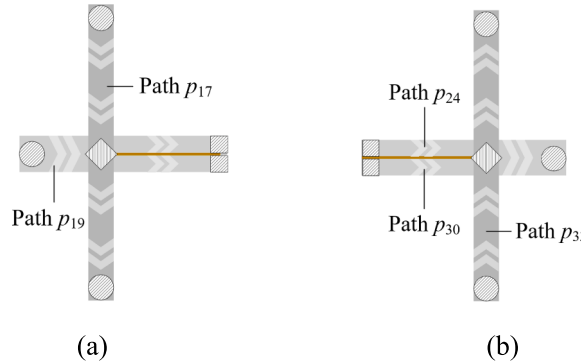


Fig. 6. Examples of path pairs for vehicle conflicts.

example, we specify that the route from HP 1 to HP 2 represents the loaded travel of an unloading container (ART task), whose path combination is $(p_2, p_1, p_6, p_{10}, p_{17}, p_{32}, p_{35})$. The route from HP 2 to HP 4 represents the empty travel of an ART arc, whose path combination is $(p_{36}, p_{37}, p_{38}, p_{39}, p_{33}, p_{18}, p_{11}, p_8, p_{10}, p_{12}, p_{13}, p_{14})$. We denote HP 3 as the handover point where an ET performs its yard operation, and its corresponding path combination is $(p_{31}, p_{19}, p_{26}, p_{27}, p_{28}, p_{29}, p_{30}, p_{25}, p_{34})$. Furthermore, to demonstrate the path pair forms resulting from vehicle conflicts between the ARTs and ETs, we present two types of path pairs in Fig. 6. Fig. 6(a) shows path pair (p_{17}, p_{19}) , whereas Fig. 6(b) shows two path pairs (p_{33}, p_{24}) and (p_{33}, p_{30}) .

As shown in Fig. 4 and Fig. 5, different types of paths have varying lengths. Paths can be categorized into the following six types based on their length. 1) Horizontal paths under cranes, i.e., paths p_1 – p_5 , p_{12} – p_{16} , p_{20} – p_{24} , p_{26} – p_{30} , and p_{35} – p_{39} in Fig. 5. 2) Vertical paths in the yard, i.e., paths p_{17} , p_{18} , and p_{31} – p_{34} in Fig. 5. 3) Vertical paths from the quayside to the yard, i.e., paths p_6 and p_7 in Fig. 5. 4) Horizontal paths from the quayside to the yard, i.e., paths p_8 and p_9 in Fig. 5. 5) Paths for ART entering the yard, i.e., paths p_{10} and p_{11} in Fig. 5. 6) Paths for ETs entering the yard block, i.e., paths p_{19} and p_{25} in Fig. 5. The values of the path lengths are presented in Section 5.1.

3.2. Basic model formulation

3.2.1. Assumptions

The following assumptions are considered in this paper without loss of generality (Lee et al., 2009; Jiang et al., 2018; Hu et al., 2019; Wang and Zeng, 2022; Zhuang et al., 2022; Li et al., 2023a).

- (1) Container storage locations in the yard and ship are known.
- (2) All containers are the same size, and the ARTs and ETs transport only one container at a time.
- (3) The travel times for the ARTs and ETs on each path are determined by the distance, and the speed is known and fixed.
- (4) The QC and ARMG scheduling schemes are predetermined and known, which determine the time widow for all containers.

3.2.2. Notations

The sets and parameters to be used in the model are described as follows.

V	set of ARTs
C^I	set of containers for ARTs. Define $C^L = C^I \cup \{k_v v \in V\}$ and $C^E = C^I \cup \{F\}$, where k_v is the unfinished task of ART v and F is a dummy termination task
C^E	set of containers for ETs
C^T	set of all containers, $C^T = C^I \cup C^E$
N	set of nodes for ARTs, $N = C^L \cup \{F\}$
A	set of arcs for ARTs, $A = \{(i, j) i \in C^L, j \in C^E, i \neq j\}$
S	set of time steps
H	set of resources, including vehicle travel paths, HPs and cranes (QCs and ARMGs)
O	set of operations, including ART travel, ET travel, ART quayside operation, ART yard side operation, ET yard operation, and vehicle occupying HP operation
O_i^L	set of vehicle travel operations for task $i \in C^T$, corresponding to the loaded travel operations for ART task $i \in C^I$ and all travel operations for ET task $i \in C^E$
O_{ij}^E	set of ART empty travel operations for arc $(i, j) \in A$
O_h	set of operations performed at resource $h \in H$
O_i	set of operations for task $i \in C^T$, where $O_i = O_i^L \cup \{o_i^F, o_i^S, o_i^A, o_i^B\}$ for task $i \in C^I$ and $O_i = O_i^L \cup \{o_i^Y, o_i^C\}$ for task $i \in C^E$
P_i	set of paths for task $i \in C^T$, corresponding to the loaded travel paths for ART task $i \in C^I$ and all travel paths for ET task $i \in C^E$
P_{ij}	set of ART empty travel paths for arc $(i, j) \in A$
P_i^A	set of path pairs (p, p') for task $i \in C^T$, indicating that paths p and p' are accessed consecutively for ART loaded travel or ET travel
P_{ij}^A	set of path pairs (p, p') for ART empty travel of arc $(i, j) \in A$
Φ	set of resource (path) pairs (h, h') where conflicts between ARTs and ETs may occur
t_p	the time for a vehicle to travel path $p \in P$
g_i^F	the time for an ART to perform the first crane operation of task $i \in C^I$
g_i^S	the time for an ART to perform the second crane operation of task $i \in C^I$
g_i^Y	the time for an ET to perform the yard operation of task $i \in C^E$
r_v	the ready time of ART $v \in V$
e_s	the beginning time of time step $s \in S$
d_h	the capacity of resource $h \in H$
$[a_i, b_i]$	[ready, due] time window of task $i \in C^T$
h_o	the resource of operation $o \in O$
p_i^0, p_i^T	the first and last paths for vehicle travel of task $i \in C^T$, corresponding to the loaded travel for ART task $i \in C^I$ and the travel before yard operation for ET task $i \in C^E$
p_{ij}^0, p_{ij}^T	the first and last paths for ART empty travel of arc $(i, j) \in A$
p_i^0, p_i^T	the first and last paths for ET travel of task $i \in C^E$ after its yard operation
o_i^F, o_i^S	the first and second crane operations of ART task $i \in C^I$
o_i^Y	the yard operation of ET task $i \in C^E$
o_i^A, o_i^B	the ART occupying HP operations of the first and second crane operations for task $i \in C^I$
o_i^C	the ET occupying HP operation of task $i \in C^E$
M	a sufficiently large positive number

The decision variables are explained as follows.

X_{vij}	binary, equals 1 if ART $v \in V$ performs task $i \in C^I$ and task $j \in C^I$ consecutively, and 0 otherwise
T_i^F	the completion time of the first crane operation for ART task $i \in C^I$
T_i^S	the completion time of the second crane operation for ART task $i \in C^I$
T_i^Y	the completion time of the yard operation for ET task $i \in C^E$
T_{ip}^L	the time for a vehicle transporting task $i \in C^T$ to leave path $p \in P_i$, corresponding to the loaded travel for ART task $i \in C^I$ and the whole travel for ET task $i \in C^E$
T_{ijp}^E	the time for an ART performing the empty travel of arc $(i, j) \in A$ to leave path $p \in P_{ij}$
D_i	the delay time of task $i \in C^T$
U_o^B	the beginning time of operation $o \in O$

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(continued)

U_o^F	the completion time of operation $o \in O$
Y_{os}	binary, equals 1 if operation $o \in O$ is performed within time step $s \in S$
Z_{os}^B	binary, equals 1 if operation $o \in O$ is performed before time step $s \in S$
Z_{os}^A	binary, equals 1 if operation $o \in O$ is performed after time step $s \in S$

3.2.3. Mathematical model

Based on the above definition, an MILP model can be constructed as follows.

$$[\mathbf{M1}] \quad \text{Min} \sum_{i \in C^I} D_i \quad (1)$$

Subject to

$$\sum_{v \in V} \sum_{(i,j) \in A: i=v} X_{vij} = 1, \forall i \in C^I \quad (2)$$

$$\sum_{(i,j) \in A: i=i} X_{vij} = \sum_{(j,i) \in A: i=i} X_{vji}, \forall i \in C^I, \forall v \in V \quad (3)$$

$$\sum_{(i,j) \in A: i=k_v} X_{vij} = 1, \forall v \in V \quad (4)$$

$$\sum_{(i,j) \in A: j=F} X_{vij} = 1, \forall v \in V \quad (5)$$

$$T_i^F \leq T_{ip_i^0}^L - t_{p_i^0}, \forall i \in C^I \quad (6)$$

$$T_{ip_i^0}^L \leq T_i^S - g_i^S, \forall i \in C^I \quad (7)$$

$$T_i^S \leq T_{ijp_{ij}^0}^E - t_{p_{ij}^0} + M \left(1 - \sum_{v \in V} X_{vij} \right), \forall (i,j) \in A \quad (8)$$

$$T_{ijp_{ij}^0}^E \leq T_j^F - g_j^F + M \left(1 - \sum_{v \in V} X_{vij} \right), \forall (i,j) \in A : j \neq F \quad (9)$$

$$T_{k_v}^S = r_v, \forall v \in V \quad (10)$$

$$a_i \leq T_i^F - g_i^F, \forall i \in C^I \quad (11)$$

$$T_i^S - b_i \leq D_i, \forall i \in C^I \quad (12)$$

$$T_{ijp}^E \leq T_{ijp^0}^E - t_{p^0} + M \left(1 - \sum_{v \in V} X_{vij} \right), \forall (i,j) \in A, \forall (p,p^0) \in P_{ij}^A \quad (13)$$

$$T_{ip}^L \leq T_{ip^0}^L - t_{p^0}, \forall i \in C^I, \forall (p,p^0) \in P_i^A \quad (14)$$

$$T_{ip_i^0}^L \leq T_i^Y - g_i^Y, \forall i \in C^E \quad (15)$$

$$T_i^Y \leq T_{ip_i^0}^L - t_{p_i^0}, \forall i \in C^E \quad (16)$$

$$a_i \leq T_{ip_i^0}^L - t_{p_i^0}, \forall i \in C^E \quad (17)$$

$$T_{ip_i^0}^L - b_i \leq D_i, \forall i \in C^E \quad (18)$$

$$\sum_{o \in O_h} Y_{os} \leq d_h, \forall h \in H, \forall s \in S \quad (19)$$

$$Y_{os} + Y_{o's} \leq 1, \forall o \in O_h, \forall o' \in O_h, \forall (h,h') \in \Phi, \forall s \in S \quad (20)$$

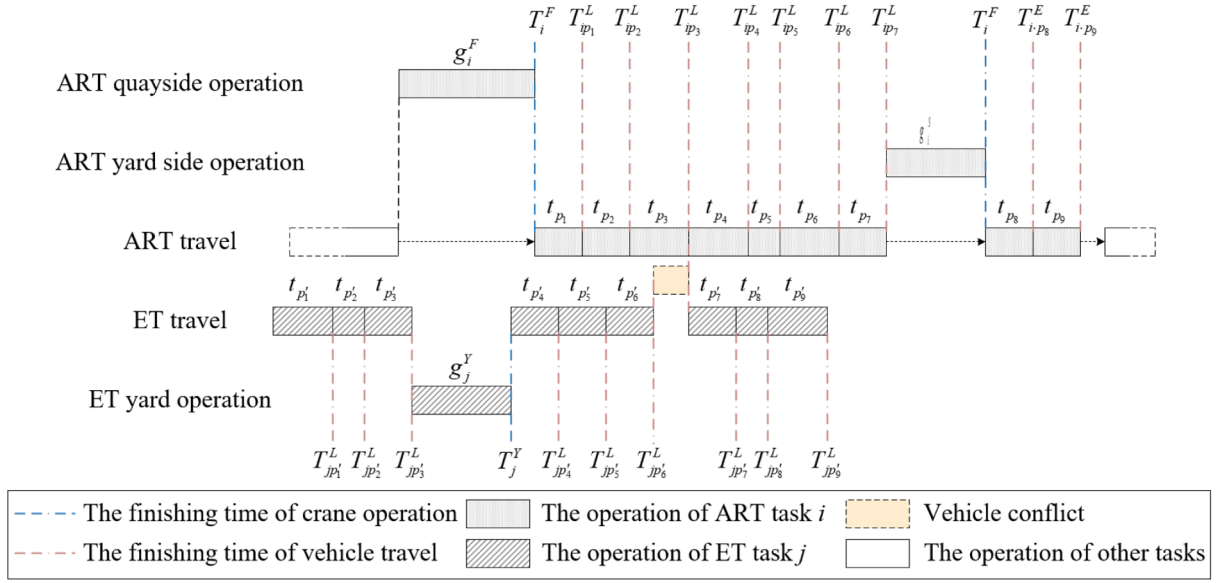


Fig. 7. An illustrative example of vehicle conflict.

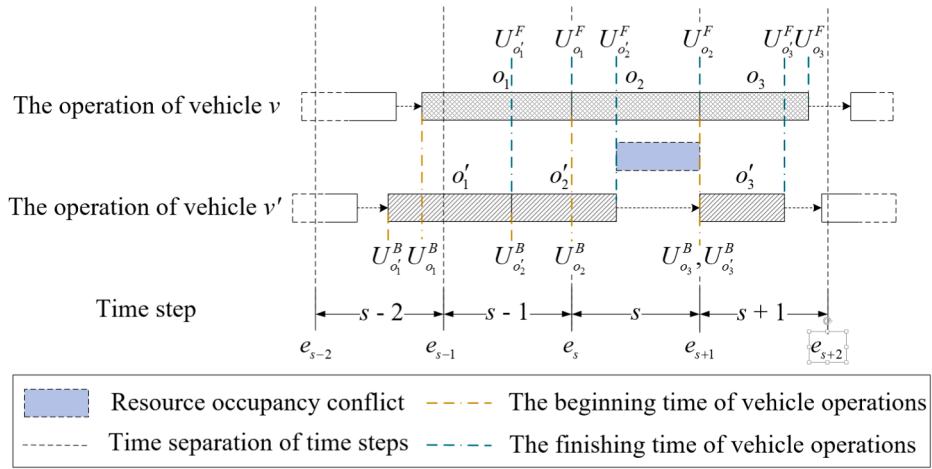


Fig. 8. The relationship between vehicle operations and time steps.

$$U_{o'}^B \leq e_{s+1} + M(1 - Y_{os}), \forall o \in O, \forall s \in S \quad (21)$$

$$e_s \leq U_{o'}^F + M(1 - Y_{os}), \forall o \in O, \forall s \in S \quad (22)$$

$$U_{o'}^F \leq e_s + M(1 - Z_{os}^B), \forall o \in O, \forall s \in S \quad (23)$$

$$e_{s+1} \leq U_{o'}^B + M(1 - Z_{os}^A), \forall o \in O, \forall s \in S \quad (24)$$

$$Y_{os} + Z_{os}^B + Z_{os}^A = 1, \forall i \in C^T, \forall o \in O_i, \forall s \in S \quad (25)$$

$$Y_{os} + Z_{os}^B + Z_{os}^A = \sum_{v \in V} X_{vij}, \forall (i, j) \in A, \forall o \in O_{ij}^E, \forall s \in S \quad (26)$$

$$U_o^B = \begin{cases} T_{ip}^L - t_p, & p = h_o, \forall i \in C^T, \forall o \in O_i^L \\ T_{ijp}^E - t_p, & p = h_o, \forall (i,j) \in A, \forall o \in O_{ij}^E \\ T_i^F - g_i^F, & o = o_i^F, \forall i \in C^I \\ T_i^S - g_i^S, & o = o_i^S, \forall i \in C^I \\ T_i^Y - g_i^Y, & o = o_i^Y, \forall i \in C^E \\ T_{ijp}^E, & o = o_j^A, \forall (i,j) \in A : j \neq F \\ T_{ip}^L, & o = o_i^B, \forall i \in C^I \text{ or } o = o_i^C, \forall i \in C^E \end{cases} \quad (27)$$

$$U_o^F = \begin{cases} T_{ip}^L, & p = h_o, \forall i \in C^T, \forall o \in O_i^L \\ T_{ijp}^E, & p = h_o, \forall (i,j) \in A, \forall o \in O_{ij}^E \\ T_i^F, & o = o_i^F, \forall i \in C^I \\ T_i^S, & o = o_i^S, \forall i \in C^I \\ T_i^Y, & o = o_i^Y, \forall i \in C^E \\ T_{ip}^L - t_{p_i^0}, & o = o_i^A, \forall i \in C^I \\ T_{ijp}^E - t_{p_{ij}^0}, & o = o_i^B, \forall (i,j) \in A : i \notin \{k_v | v \in V\} \\ T_{ip}^L - t_{p_i^0}, & o = o_i^C, \forall i \in C^E \end{cases} \quad (28)$$

$$X_{vij} \in \{0, 1\}, \forall v \in V, \forall (i,j) \in A \quad (29)$$

$$Y_{os}, Z_{os}^B, Z_{os}^A \in \{0, 1\}, \forall o \in O, \forall s \in S \quad (30)$$

$$T_{ip}^L, D_i \geq 0, \forall i \in C^T, \forall p \in P_i \quad (31)$$

$$T_i^F, T_i^S \geq 0, \forall i \in C^I \quad (32)$$

$$T_{ijp}^E \geq 0, \forall (i,j) \in A, \forall p \in P_{ij} \quad (33)$$

$$U_o^B, U_o^F \geq 0, \forall o \in O \quad (34)$$

$$T_i^Y \geq 0, \forall i \in C^E \quad (35)$$

The objective function (1) minimizes the total delay time, including the delay time of the ART tasks and the ET delay time. The constraints of the model are classified into the following five categories.

- **ART task assignment and sequence constraints.** Constraint (2) indicates that each ART task is performed by exactly one ART. Constraint (3) ensures the continuity of the ART transportation, indicating that each ART has one predecessor and one successor task. Constraint (4) indicates that each ART begins from its unfinished task, whereas constraint (5) ensures that each ART ends on a dummy task.
- **Time relationship constraints for ART transportation.** Constraint (6) guarantees a time relationship between the first crane operation and loaded travel for ART tasks, where the beginning time of ART loaded travel is not less than the completion time of the first crane operation. Constraint (7) indicates a time relationship between the second crane operation and the loaded travel for ART tasks. Constraints (8) and (9) ensure a time relationship between crane operations and ART empty travel if arc (i, j) is accessed. A virtual path with zero transport time is defined for an arc with zero empty travel distance. Constraint (10) defines the ART ready time. Constraint (11) indicates that the beginning time of an ART task is not earlier than its ready time. Constraint (12) defines the delay times of the ART tasks. Constraint (13) describes the time relationship of the ART empty travel operations if arc (i, j) is accessed, where the time to enter path p is not less than the time to leave path p . For ART tasks, constraint (14) defines the completion times of the ART loaded travel operations.
- **Time relationship constraints for ET transportation.** For ET tasks, constraint (14) defines the time relationship between all ET travel operations. Constraints (15) and (16) ensure a time relationship between the ET yard operation and travel operations. Constraint (17) indicates that the beginning time of ET operations is not earlier than the ET ready time. Constraint (18) defines the delay times of the ET tasks.
- **Resource capacity and vehicle conflict constraints.** Constraints (19) and (20) represent the core constraints of this group. Constraint (19) defines the resource capacity, indicating that the number of vehicles for each resource (path, HP, and crane) at each

time step cannot exceed its capacity. Constraint (20) avoids vehicle conflicts at ART and ET intersections (see Fig. 5). Constraints (21)–(24) ensure the time relationship between operations and time steps, where constraints (21) and (22) correspond to operations performed within time step s , constraints (23) corresponds to operations performed before time step s , and constraints (24) corresponds to operations performed after time step s . Constraint (25) indicates that all operations of task i must be performed before, within, or after time step s . Constraint (26) is similar to constraint (25), but for the ART empty travel operations. Constraints (27) and (28) define the beginning and completion times of all operations, respectively.

- **Constraints defining variables.** Constraints (29)–(35) illustrate the ranges of decision variables.

To provide a clearer description of the modeling concepts, we present an illustrative example in Fig. 7 and depict the relationship between vehicle operations and time steps in Fig. 8, where the vehicle conflict between ART and ET occurs in Fig. 7 and the resource occupancy conflict arises in Fig. 8 because of limited resource capacity. Fig. 7 displays all the operations of ART task i and ET task j , with the time points of each operation marked. ART task i is an unloading container whose loaded travel paths are $(p_1, p_2, p_3, p_4, p_5, p_6, p_7)$, and the empty travel paths from ART task i to the next task are (p_8, p_9) . The travel paths of ET task j are $(p'_1, p'_2, p'_3, p'_4, p'_5, p'_6, p'_7, p'_8, p'_9)$, where (p'_1, p'_2, p'_3) corresponds to travel before yard operation and $(p'_4, p'_5, p'_6, p'_7, p'_8, p'_9)$ corresponds to travel after yard operation. In Fig. 7, a vehicle conflict occurs between the ART traveling on path p_3 and the ET traveling on path p'_7 , indicating that $(p_3, p'_7) \in \Phi$. As depicted in Fig. 8, vehicle v performs operations o_1 , o_2 , and o_3 , whereas vehicle v' performs operations o'_1 , o'_2 , and o'_3 . The start and completion times of all operations and the time intervals of the time steps are marked. Note that vehicles $v, v' \in V \cup C^E$, where $v \in V$ indicates that vehicle v is an ART and $v \in C^E$ indicates that vehicle v is an ET. From Fig. 8, it is evident that operations o_2 and o'_3 are performed in the same resource ($h_{o_2} = h_{o'_3}$), and the number of vehicles accommodated in resource h_{o_2} reaches its capacity at time step s . Therefore, operation o'_3 must wait until operation o_2 is complete. It should be emphasized that the types of resource occupancy conflicts include the following: 1) Different ARTs occupying the same ART path or HP; 2) Different ETs occupying the same ET path or HP; 3) Different ARTs occupying the same QC; and 4) Different ARTs and ETs occupying the same ARMG.

In model **M1**, constraints (2)–(14) ensure the transportation of internal trucks (ARTs), while constraints (14)–(18) ensure the operations of ETs. The interaction between ARTs and ETs is ensured by constraint (20), and the core constraint of resource capacity limitation is constraint (19). Constraints (21)–(28) represent the time relationship between vehicle operations and time steps. Based on the above analysis, model **M1** can be adapted to schedule internal and external trucks with different layouts. First, for the perpendicular-layout ACT where the transportation areas of internal and external trucks are physically separated, constraints (2)–(19) and (21)–(35) can be retained, and constraint (20) is removed. In this layout, the travel of internal trucks and ETs is two relatively independent processes, and there is no intersection between them. Note that although the model changes are minor, adjustments are needed to the sets based on the layout, including resource-related sets (H , Φ), path-related sets (P_i , P_{ij} , P_i^A , P_{ij}^A), operation-related sets (O , O_i^L , O_{ij}^E , O_h , O_i), and related parameters (d_h , h_o). Second, for the parallel-layout container terminal with lanes on only one side of the yard block, all constraints of model **M1** are retained. The main difference of this layout from the layout we studied is that internal and external trucks in this layout travel on the same side of the yard block. For the layout with lanes on both sides of the yard block, the path sets of internal and external trucks in the block are completely different. For the layout with lanes on one side of the block, internal and external trucks have the same path sets in the block. Therefore, although the model remains unchanged, the number of paths in this layout is greatly reduced. Correspondingly, the related sets and parameters also need to be changed according to the path structure. In summary, when applying model **M1** to schedule internal and external trucks with different layouts, the most crucial changes are to adjust sets and parameters based on the path structure.

3.3. Complexity of the problem

The MILP model constructed above for the AESP can be solved directly using the commercial solver Gurobi. However, due to the large number of constraints and variables in the model, obtaining optimal solutions for medium- and large-scale instances is extremely time-consuming. Specifically, the model of the AESP contains $|V| \cdot |A| + 2 \cdot |C^T| + |C^L| + \sum_{i \in C^T} |P_i| + \sum_{(i,j) \in A} |P_{ij}| + (3 \cdot |S| + 2) \cdot |O|$ variables and $9 \cdot |C^T| + (4 \cdot |O| + |H|) \cdot |S| + \sum_{i \in C^T} (2 \cdot |O_i^L| + |S| \cdot |O_i| + |P_i^A|) + \sum_{(i,j) \in A} (2 \cdot |O_{ij}^E| + |S| \cdot |O_{ij}^E| + |P_{ij}^A|) + \sum_{(h,h') \in \Phi} |O_h| \cdot |O_{h'}| \cdot |S| + 8 \cdot |C^E| + 4 \cdot |A|$ constraints. Even for the small-scale instance of 4 ARTs, 10 ART tasks, 10 ET tasks, and 100 time steps with the layout shown in Fig. 5, model **M1** includes more than 500,000 decision variables and 1,400,000 constraints. In addition, without considering ET operation and limited resource capacity, the AESP is a classical vehicle routing problem (VRP) that has been proven to be NP-hard. Therefore, the AESP is also an NP-hard problem.

4. Solution approach

Model **M1** contains a considerable number of variables and constraints as described in Section 3.3, making it computationally intractable even for small-scale instances with state-of-the-art MIP solvers such as Gurobi. Therefore, we design a branch-and-price heuristic (BPH) algorithm that adopts the structure of the branch-and-bound method and iteratively invokes the column generation method (Liu et al., 2023a; Meng et al., 2015). The solution approach is inspired by the fact that the AESP includes two types of vehicles (ART and ET), and the operation process of each vehicle can be regarded as a route. Generating all routes simultaneously in a single

model is computationally challenging and unrealistic. However, the problem can be significantly simplified if the relationship between different vehicles/routes is described in terms of vehicle occupation resources. Accordingly, the operational processes of each vehicle can be scheduled independently. The framework of the solution algorithm is presented in [Section 4.1](#), and the details of the embedded sub-processes are provided in [Sections 4.2–4.5](#).

4.1. Framework of the proposed BPH algorithm

The main idea of the BPH algorithm is to first reformulate model **M1** into a route-based model (master problem, **MP**), as described in [Section 4.2](#). To reduce the computational burden, we utilize the column generation method to generate a finite number of high-quality routes/columns to solve the restricted master problem (**RMP**), rather than enumerating all possible routes/columns. The column generation method is embedded into a branch-and-bound structure to construct the BPH algorithm and generate feasible integer solutions. Several acceleration methods are proposed to improve this algorithm. Motivated by the classic branch-and-price algorithm framework, the proposed BPH algorithm applies the master problem, sub-problem, initial solution construction algorithm, branching strategy, and acceleration techniques adapted to the problem characteristics. To balance the computational time and solution quality, the number of nodes in the BPH algorithm is limited to B^{node} , and the column generation of each node is terminated early ([Section 4.5.2](#)), rendering the algorithm heuristic. The algorithm framework is presented in two parts: ease of understanding and conciseness. The BPH algorithm framework is described in Algorithm 1, and the column generation steps are introduced in Algorithm 2, where the **MP** and **RMP** are solved using Gurobi.

(continued on next page)

(continued)

Algorithm 1 Branch-and-price heuristics algorithm**Input:** data file, node limit B^{node} **Output:** final feasible solution Ξ^F and its objective value D^F

```

01: initialize  $LB = 0$ ,  $UB \leftarrow \infty$ ,  $\Omega^{BP} \leftarrow \emptyset$ ,  $\Omega^{CG} \leftarrow \emptyset$ ,  $\Upsilon = \{0\}$ ,  $\delta^{BP} = 0$ 
02: obtain  $\Xi^{RMP}$ ,  $D^{RMP}$ , and  $\Pi^{RMP}$  based on the column generation in Algorithm 2
03:  $LB = D^{RMP}$ ,  $LB'_0 = D^{RMP}$ ,  $\Omega^{CG} \leftarrow$  the columns with non-zero decision variable values
04: solve the MP based on  $\Pi^{RMP}$  to generate solution  $\Xi^{MP}$  and its objective value  $D^{MP}$ 
05:  $UB = D^{MP}$ ,  $\Omega^{CG} \leftarrow$  the columns with decision variable values equal to 1
06: do
07:   if  $LB \geq UB$  or  $\Upsilon = \emptyset$  then # stop criterion
08:      $\Xi^F = \Xi^{MP}$ ,  $D^F = D^{MP}$ ,  $\delta^{BP} = 1$ 
09:   else if the number of nodes  $\geq B^{node}$  then # stop criterion
10:     solve the MP based on  $\Omega^{BP} \cup \Pi^{RMP}$  to generate  $\Xi^{MP}$  and  $D^{MP}$ 
11:      $\Xi^F = \Xi^{MP}$ ,  $D^F = D^{MP}$ ,  $\delta^{BP} = 1$ 
12:   else #node exploration
13:     select node  $\theta^*$  from  $\Upsilon$  with the node selection strategy presented in Section 4.4
14:     branch node  $\theta^*$  into two nodes  $\theta'_1$  and  $\theta'_2$  by the branching strategy in Section 4.4
15:      $\Upsilon = \Upsilon \setminus \{\theta^*\}$  #remove node  $\theta^*$  from node set
16:     for  $\theta \in \{\theta'_1, \theta'_2\}$  do
17:       solve the RMP of  $\theta$  by rows 2-26 of Algorithm 2 to obtain  $\Xi^{RMP}$  and  $D^{RMP}$ 
18:        $LB'_\theta = D^{RMP}$ ,  $\Omega^{BP} \leftarrow$  the columns with non-zero decision variable values
19:       if  $\Xi^{RMP}$  is an integer solution and  $D^{RMP} < UB$  then
20:          $\Xi^{MP} = \Xi^{RMP}$ ,  $D^{MP} = D^{RMP}$ ,  $UB = D^{RMP}$ 
21:       else if  $D^{RMP} < UB$  then
22:         solve the MP with  $\Omega^{CG}$  and columns just obtained to generate  $\Xi^{MP}$  and  $D^{MP}$ 
23:          $UB = \min\{UB, D^{MP}\}$ ,  $\Upsilon = \Upsilon \cup \{\theta\}$ ,  $\Omega^{BP} \leftarrow$  the columns with values equal to 1
24:       end if
25:     end for
26:      $\Upsilon = \Upsilon \setminus \{\theta \in \Upsilon \mid LB'_\theta \geq UB\}$ 
27:      $LB = \min_{\theta \in \Upsilon} \{LB'_\theta\}$  #update the global lower bound
28:   end if
29: loop until  $\delta^{BP} = 1$ 
30: return  $\Xi^F$  and  $D^F$ 

```

In **Algorithm 1**, row 1 initializes global lower bound LB , upper bound UB , node set Υ , and termination parameter δ^{BP} . Rows 2–5 obtain lower bound LB'_0 of the root node, LB , and UB . Two column sets Ω^{BP} and Ω^{CG} are defined, which contain the columns with non-zero decision variable values during the algorithm iterations. Accordingly, the **MP** is solved repeatedly at each node based on set Ω^{CG} and newly obtained columns (rows 22–23), which have been proven to effectively improve the upper bound (Liu et al., 2023a). When the number of nodes reaches node limit B^{node} , the **MP** is directly solved based on the columns in set Ω^{BP} and those generated at the root node, resulting in a high-quality final feasible solution (rows 9–11). In the node exploration process (rows 13–27), rows 13–14 select a node and branch it into two child nodes, while the computation of each child node is conducted in rows 17–23 to obtain lower bound

LB_θ of node θ and update upper bound UB . Note that row 26 removes nodes whose lower bound exceeds the upper bound.

Algorithm 2 Column generation

Input: data file, sub-problem set P^{sub}

Output: RMP solution Ξ^{RMP} , objective value D^{RMP} , and column set Π^{RMP}

```

01: obtain initial column set  $\Pi^{RMP}$  based on the algorithm presented in Section 4.3.3
02: do
03:   solve the RMP based on  $\Pi^{RMP}$  to generate dual variables,  $\Xi^{RMP}$ , and  $D^{RMP}$ 
04:    $\delta'_\tau \leftarrow 1, \forall \tau \in P^{sub}$  #initialize the termination parameters of sub-problems
05:   for  $\tau \in P^{sub}$  do
06:     if the sum of all dual variables for sub-problem  $\tau \leq 0$  then
07:        $\delta'_\tau = 0$ 
08:       continue
09:     end if
10:     if  $\tau \leq |V|$  then #ART sub-problem
11:       solve sub-problem  $\tau$  based on the algorithm proposed in Section 4.5.1
12:     else #ET sub-problem
13:       solve sub-problem  $\tau$  directly using Gurobi
14:     end if
15:     generate set  $\Pi^{sub}$  containing at most  $Col$  columns with negative reduced costs
16:     if  $\Pi^{sub} = \emptyset$  then
17:        $\delta'_\tau = 0$ 
18:     end if
19:      $\Pi^{RMP} = \Pi^{RMP} \cup \Pi^{sub}$ 
20:   end for
21:    $\delta = \sum_{\tau \in P^{sub}} \delta'_\tau$  #the termination parameter of the column generation
22:   if meet the termination criteria described in Section 4.5.2 then
23:      $\delta = 0$ 
24:   end if
25: loop until  $\delta = 0$ 
26: return  $\Xi^{RMP}$ ,  $D^{RMP}$ , and  $\Pi^{RMP}$ 

```

In Algorithm 2, the column generation iteratively solves the RMP with column set Π^{RMP} to obtain dual variables (row 3), which are then transferred into sub-problems to generate columns with negative reduced costs (rows 5–20). The AESP consists of $|V|$ ART sub-problems (Section 4.3.1) and $|C^E|$ ET sub-problems (Section 4.3.2), and the sub-problem set can thus be defined as $P^{sub} = \{1, 2, \dots, |V| + |C^E|\}$. Note that a dual variable acceleration method (Section 4.5.3) is designed in rows 6–9 to disregard sub-problems that cannot generate columns with negative reduced costs. Row 15 employs the multi-column addition method described in Section 4.5.2 to add multiple columns for solving the RMP. The column generation process is repeated until no new columns can be obtained (rows 16–18) or the termination condition is reached (rows 22–24).

4.2. Route-based model

In model **M1**, numerous constraints guarantee the temporal relationship between vehicle operations and between time steps and vehicle operations (constraints (6)–(18) and (21)–(28)). These constraints can be effectively addressed by reconstruction (Wang and Sheu, 2019), where the operation-related constraints are considered in the pricing sub-problems. Accordingly, we reformulate the AESP into a route-based model based on all ART routes and ET routes, with the objective of obtaining the optimal route subset. In the route-based model, R is defined as the set of all routes, consisting of ART route set R^I and ET route set R^E . We define parameter w_r as the total delay time of route $r \in R$. Let binary coefficient α_{ir} be 1 if task $i \in C^T$ is conducted in route $r \in R$, and 0 otherwise. Let binary

coefficient β_{vr} be 1 if route $r \in R^I$ corresponds to ART $v \in V$, and 0 otherwise. Let binary coefficient ε_{hsr} be 1 if the vehicle of route $r \in R$ occupies resource $h \in H$ in time step $s \in S$, and 0 otherwise. Specifically, we define a decision variable λ_r , which indicates if route $r \in R$ is incorporated in the final solution. The route-based model (**MP**) can thus be formulated as follows.

$$[\mathbf{MP}] \quad \text{Min} \sum_{r \in R} w_r \lambda_r \quad (36)$$

Subject to

$$\sum_{r \in R} \alpha_{ir} \lambda_r = 1, \forall i \in C^T \quad (37)$$

$$\sum_{r \in R^I} \beta_{vr} \lambda_r = 1, \forall v \in V \quad (38)$$

$$\sum_{r \in R} \varepsilon_{hsr} \lambda_r \leq d_h, \forall h \in H, \forall s \in S \quad (39)$$

$$\varepsilon_{hsr} \lambda_r + \varepsilon'_{hsr} \lambda_{r'} \leq 1, \forall r \in R^I, \forall r' \in R^E, \forall (h, h') \in \Phi, \forall s \in S \quad (40)$$

$$\lambda_r \in \{0, 1\}, \forall r \in R \quad (41)$$

The objective function (36) minimizes the total delay time. Constraint (37) ensures that each task is executed only once. Constraint (38) states that only one route can be selected for each ART. Constraint (39) indicates that the number of vehicles for each resource cannot exceed its capacity at any time step. Constraint (40) prevents vehicle conflicts between the ART and ET routes. Constraint (41) specifies the range of decision variable λ_r . Constraint (40) corresponds to constraint (20) of model **M1**, where **M1** and **MP** have $\sum_{(h,h') \in \Phi} |O_h| \cdot |O_{h'}| \cdot |S|$ and $|R^I| \cdot |R^E| \cdot |\Phi| \cdot |S|$ constraints, respectively. Therefore, the number of constraints is effectively reduced by reformulating model **M1** based on $|O_h| \cdot |O_{h'}| \gg |R^I| \cdot |R^E|$.

4.3. Column generation

The number of feasible routes in set R exponentially increases as the problem size increases, resulting in an explosion of CPU time. In fact, only a small number of columns is incorporated into the optimal solution. Consequently, to ensure a reasonable computational time, we design a column generation that is repeatedly invoked in the BPH algorithm to generate a relatively small set of high-quality columns. In the column generation process, a linear programming relaxation of the **MP**, the restricted master problem (**RMP**), is proposed based on a subset of route set R . The dual variables obtained by solving the **RMP** are then input into the pricing sub-problems to generate promising columns. Specifically, there are two types of sub-problems: ART sub-problems and ET sub-problems. The dual variables for constraints (37)–(40) are denoted μ_i , π_v , η_{hs} , and $\chi'_{rr'hh's}$, respectively, and define $\chi_{hh's} = \sum_{r \in R^I} \sum_{r' \in R^E} \chi'_{rr'hh's}$.

4.3.1. The pricing sub-problems for ARTs

For each column generation iteration, there are $|V|$ ART sub-problems, where index v is removed as each sub-problem is solved independently. The reduced cost c_r for the ART route is given by equation (42).

$$c_r = w_r - \sum_{i \in C^I} \mu_i \alpha_{ir} - \sum_{h \in H} \sum_{s \in S} \eta_{hs} \varepsilon_{hsr} - \sum_{(h,h') \in \Phi} \sum_{s \in S} \chi_{hh's} \varepsilon_{hsr} - \pi, \forall r \in R^I \quad (42)$$

To represent the reduced cost of an ART route in terms of arcs, we introduce two new variables x_{ij} and $\bar{e}_{hsij}$, where x_{ij} equals 1 if arc $(i, j) \in A$ is visited by the route (and 0 otherwise), and $\bar{e}_{hsij}$ equals 1 if arc $(i, j) \in A$ is visited and occupies resource $h \in H$ in time step $s \in S$ (and 0 otherwise). Thus, the reduced cost c_{ij} of arc (i, j) can be specified using (43).

$$c_{ij} = \begin{cases} D_j - \mu_i x_{ij} - \sum_{h \in H} \sum_{s \in S} \eta_{hs} \bar{e}_{hsij} - \sum_{(h,h') \in \Phi} \sum_{s \in S} \chi_{hh's} \bar{e}_{hsij}, & j \in C^I \\ -\pi, & j \in F \end{cases} \quad (43)$$

The pricing sub-problem for an ART can be formulated as follows.

$$[\mathbf{SP_1}] \quad \text{Min} \sum_{(i,j) \in A} c_{ij} \quad (44)$$

Subject to constraints (21)–(24), (30), (32)–(34), and

$$\sum_{(i,j) \in A: i=i} x_{ij} = \sum_{(j,i) \in A: i=i} x_{ji}, \forall i \in C^I \quad (45)$$

$$\sum_{(i,j) \in A: i=k} x_{ij} = 1 \quad (46)$$

$$\sum_{(i,j) \in A: j=F} x_{ij} = 1 \quad (47)$$

$$T_i^F \leq T_{ip_i^0}^L - t_{p_i^0} + M \left(1 - \sum_{(i,j) \in A: i=i} x_{ij} \right), \forall i \in C^I \quad (48)$$

$$T_{ip_i^T}^L \leq T_i^S - g_i^S + M \left(1 - \sum_{(i,j) \in A: i=i} x_{ij} \right), \forall i \in C^I \quad (49)$$

$$T_i^S \leq T_{ijp_{ij}^0}^E - t_{p_{ij}^0} + M(1 - x_{ij}), \forall (i,j) \in A \quad (50)$$

$$T_{ijp_{ij}^T}^E \leq T_j^F - g_j^F + M(1 - x_{ij}), \forall (i,j) \in A \quad (51)$$

$$T_k^S = r \quad (52)$$

$$a_i \leq T_i^F - g_i^F + M \left(1 - \sum_{(i,j) \in A: i=i} x_{ij} \right), \forall i \in C^I \quad (53)$$

$$T_i^S - b_i \leq D_i + M \left(1 - \sum_{(i,j) \in A: i=i} x_{ij} \right), \forall i \in C^I \quad (54)$$

$$T_{ijp}^E \leq T_{ijp}^E - t_p + M(1 - x_{ij}), \forall (i,j) \in A, \forall (p,p') \in P_{ij}^A \quad (55)$$

$$T_{ip}^L \leq T_{ip}^L - t_p + M \left(1 - \sum_{(i,j) \in A: i=i} x_{ij} \right), \forall i \in C^I, \forall (p,p') \in P_i^A \quad (56)$$

$$Y_{os} + Z_{os}^B + Z_{os}^A = x_{ij}, \forall (i,j) \in A, \forall o \in O_j \cup O_{ij}^E, \forall s \in S \quad (57)$$

$$\bar{e}_{hsij} = \sum_{o \in (O_j \cup O_{ij}^E) \cap O_h} Y_{os}, \forall (i,j) \in A, \forall h \in H, \forall s \in S \quad (58)$$

$$U_o^B = \begin{cases} T_{ip}^L - t_p, & p = h_o, \forall i \in C^I, \forall o \in O_i^L \\ T_{ijp}^E - t_p, & p = h_o, \forall (i,j) \in A, \forall o \in O_{ij}^E \\ T_i^F - g_i^F, & o = o_i^F, \forall i \in C^I \\ T_i^S - g_i^S, & o = o_i^S, \forall i \in C^I \\ T_{ijp_{ij}^T}^E, & o = o_j^A, \forall (i,j) \in A : j \neq F \\ T_{ip_i^T}^L, & o = o_i^B, \forall i \in C^I \end{cases} \quad (59)$$

$$U_o^F = \begin{cases} T_{ip}^L, & p = h_o, \forall i \in C^I, \forall o \in O_i^L \\ T_{ijp}^E, & p = h_o, \forall (i,j) \in A, \forall o \in O_{ij}^E \\ T_i^F, & o = o_i^F, \forall i \in C^I \\ T_i^S, & o = o_i^S, \forall i \in C^I \\ T_{ip_i^0}^L - t_{p_i^0}, & o = o_i^A, \forall i \in C^I \\ T_{ijp_{ij}^0}^E - t_{p_{ij}^0}, & o = o_i^B, \forall (i,j) \in A : i \neq k \end{cases} \quad (60)$$

$$x_{ij}, \bar{e}_{hsij} \in \{0, 1\}, \forall (i,j) \in A, \forall h \in H, \forall s \in S \quad (61)$$

$$T_{ip}^L, D_i \geq 0, \forall i \in C^I, \forall p \in P_i \quad (62)$$

The objective function (44) minimizes the reduced cost of the ART route. Constraints (45)–(47) are equivalent to constraints (3)–(5), namely the ART task assignment and sequence constraints. Constraints (48)–(51) and (53)–(56) are copies of constraints (6)–(9) and (11)–(14), respectively, which ensure the time relationships between ART operations if arc (i, j) is accessed. Constraint (52) defines the beginning time of ART, where k and r represent the unfinished task and ready time, respectively, with the ART index v removed. Constraint (57) states that if arc (i, j) is accessed by the ART, the operations of task j and the empty travel operations of arc (i, j) must be performed before, within, or after time step s . Constraint (58) indicates that variable $\bar{e}_{hsij}$ equals 1 if the operations in set $O_j \cup O_{ij}^E$ occupy resource h in time step s . Constraints (59) and (60) are equivalent to constraints (27) and (28) but only for ART operations. Constraints (61) and (62) define the domains of decision variables.

4.3.2. The pricing sub-problems for ETs

The AESP involves $|C^E|$ ET sub-problems, each of which corresponds to an ET. Moreover, each ET sub-problem conducts only one ET task, implying that index i is fixed in the sub-problem model. The reduced cost c_r of an ET route can be formulated using equation (63).

$$c_r = D_i - \sum_{i \in C^E} \mu_i \alpha_{ir} - \sum_{h \in H} \sum_{s \in S} \eta_{hs} \varepsilon_{hsr} - \sum_{(h,h') \in \Phi} \sum_{s \in S} \chi_{hh's} \varepsilon_{h'sr}, \forall r \in R^E \quad (63)$$

Let the variable $\bar{e}_{hsi}$ be 1 if ET task $i \in C^E$ occupies resource $h \in H$ in time step $s \in S$, and 0 otherwise. Thus, the pricing sub-problem for an ET can thus be formulated as follows.

$$[\text{SP-2}] \quad \text{Min } D_i - \sum_{h \in H} \sum_{s \in S} \eta_{hs} \bar{e}_{hsi} - \sum_{(h,h') \in \Phi} \sum_{s \in S} \chi_{hh's} \bar{e}_{h'si} - \mu_i \quad (64)$$

Subject to constraints (21)–(24) and

$$T_{ip}^L \leq T_{ip}^L - t_p, \forall (p, p') \in P_i^A \quad (65)$$

$$T_{ip_i^T}^L \leq T_i^Y - g_i^Y \quad (66)$$

$$T_i^Y \leq T_{ip_i^O}^L - t_{p_i^O} \quad (67)$$

$$a_i \leq T_{ip_i^O}^L - t_{p_i^O} \quad (68)$$

$$T_{ip_i^P}^L - b_i \leq D_i \quad (69)$$

$$Y_{os} + Z_{os}^B + Z_{os}^A = 1, \forall o \in O_i, \forall s \in S \quad (70)$$

$$\bar{e}_{hsi} = \sum_{o \in O_i \cap O_h} Y_{os}, \forall h \in H, \forall s \in S \quad (71)$$

$$U_o^B = \begin{cases} T_{ip}^L - t_p, & p = h_o, \forall o \in O_i^L \\ T_i^Y - g_i^Y, & o = o_i^Y \\ T_{ip_i^T}^L, & o = o_i^C \end{cases} \quad (72)$$

$$U_o^F = \begin{cases} T_{ip}^L, & p = h_o, \forall o \in O_i^L \\ T_i^Y, & o = o_i^Y \\ T_{ip_i^O}^L - t_{p_i^O}, & o = o_i^C \end{cases} \quad (73)$$

$$\bar{e}_{hsi}, Y_{os}, Z_{os}^B, Z_{os}^A \in \{0, 1\}, \forall h \in H, \forall s \in S, \forall o \in O_i \quad (74)$$

$$T_{ip}^L, D_i, T_i^Y, U_o^B, U_o^F \geq 0, \forall p \in P_i, \forall o \in O_i \quad (75)$$

The objective function (64) is to minimize the reduced cost of the ET route. Constraints (65)–(69) replicate constraints (14)–(18) with a fixed ET task i . Constraints (70) and (71) are similar to constraints (57) and (58), where constraint (70) ensures the relationship between the operations of ET task i and time step s , and constraint (71) defines the value of variable $\bar{e}_{hsi}$. Constraints (72) and (73) are equivalent to constraints (27) and (28) but apply to ET operations. Constraints (74) and (75) indicate the domains of decision variables.

4.3.3. The construction of initial feasible solution

The construction of the initial feasible solution has a significant effect on the convergence speed of the column generation. A high-quality initial solution can effectively reduce the computational load of the BPH algorithm. However, the complexity of decision-making in the AESP, which requires providing ART and ET scheduling schemes considering limited resource capacity and vehicle conflict, makes obtaining a feasible solution challenging. Therefore, we propose an initial solution construction heuristic algorithm to generate a series of feasible columns to execute column generation, where the feasible columns contain the ART and ET routes, as presented in Algorithm 3.

Algorithm 3 Initial solution construction algorithm

Input: data file, C^T , V

Output: initial feasible solution Θ

```

01: initialize resource occupancy information  $\Lambda$ ,  $\Theta \leftarrow \emptyset$ ,  $C' \leftarrow C^T$ 
02: while  $C' \neq \emptyset$  do
03:    $\Psi_1 \leftarrow \emptyset$ 
04:   for  $i \in C'$  do
05:     if  $i$  is an ART task then
06:       for  $v \in V$  do
07:         assign  $i$  to  $v$  considering resource capacity and vehicle conflict based on  $\Lambda$ 
08:         obtain total delay time  $\Gamma_1(i, v)$  and makespan  $\Gamma_2(i, v)$ 
09:          $\Psi_1 = \Psi_1 \cup \{(i, v)\}$ 
10:       end for
11:     else #  $i$  is an ET task
12:       conduct ET task  $i$  considering resource capacity and vehicle conflict based on  $\Lambda$ 
13:       obtain total delay time  $\Gamma_1(i, i)$  and makespan  $\Gamma_2(i, i)$ 
14:        $\Psi_1 = \Psi_1 \cup \{(i, i)\}$ 
15:     end if
16:   end for
17:    $(i', v') = \arg \min_{(i, v) \in \Psi_1} \Gamma_1(i, v)$ 
18:   generate set  $\Psi_2 = \{(i, v) \in \Psi_1 \mid \Gamma_1(i, v) = \Gamma_1(i', v')\}$ 
19:    $(i^*, v^*) = \arg \min_{(i, v) \in \Psi_2} \Gamma_2(i, v)$ 
20:   if  $i^*$  is an ART task then
21:     conduct the operations of task  $i^*$  by ART  $v^*$  in sequence based on  $\Lambda$ 
22:     update resource occupancy information  $\Lambda$ 
23:   else
24:     conduct the operations of ET task  $i^*$  in sequence based on  $\Lambda$ 
25:     update resource occupancy information  $\Lambda$ 
26:   end if
27:    $\Theta = \Theta \cup \{(i^*, v^*, \Lambda)\}$ ,  $C' = C' \setminus \{i^*\}$ 
28: end while
29: return  $\Theta$ 

```

The main idea of Algorithm 3 is to decide the vehicle and task assignment scheme based on the total delay time and makespan, and to schedule the execution sequence of conflicting operations according to first-come first-service rule. The details of the algorithm are described as follows. Rows 3–19 are designed to find the vehicle and task combination (i^*, v^*) , while rows 20–26 are developed to execute the operations of the combination in sequence. Considering limited resource capacity and vehicle conflict, rows 7–8 and 12–13

obtain the total delay time and makespan of the vehicle and task combination based on current resource occupancy information Λ . Note that each ET task corresponds to precisely one ET; thus, the ET index in row 13 is represented by i . Rows 17–19 first filter out a set of vehicle and task combinations depending on the total delay time, and then select the combination with the minimum makespan. In row 21, the ART empty travel operation, first crane operation, loaded travel operation, and second crane operation are performed in sequence considering limited resource capacity and vehicle conflict. Row 24 is similar to row 21, where the ET travel before yard operation, yard operation, and travel after yard operation are conducted in sequence. The first-come first-service rule is applied in rows 21 and 24, that is, the operations performed first have priority in occupying resources, and the subsequent operations must avoid the resource occupancy conflict and vehicle conflict with current information Λ . The occupancy information of vehicle paths, HPs, and cranes is updated in rows 22 and 25.

4.4. Branching and node selection strategies

The solution obtained from the **RMP** is generally a non-integer solution because we solve the linear relaxation of the **MP**. To generate an integer solution of the AESP, we incorporate the column generation into a search tree framework known as the branch-and-bound framework, where the column generation process needs to be repeated at each node. In this regard, we design branching and node selection strategies for the search tree (Liu et al., 2023a). During the branching process, if the values of variable λ_r for some routes are fractional, we prioritize selecting route r^* with a variable value closest to 0.5. In cases where multiple routes have the same variable value, the route with the largest delay time is selected. Two child nodes can thus be generated as follows. For the first child node, we set variable λ_{r^*} to 1, remove the sub-problem corresponding to route r^* , and delete the routes/columns in the **RMP** that have the same vehicle ($r \in \{r | \beta_{vr} = 1, \beta_{vr^*} = 1, \forall v \in V\} \setminus \{r^*\}$) and the same containers ($r \in \{r | \alpha_{ir} = 1, \alpha_{ir^*} = 1, \forall i \in C^T\} \setminus \{r^*\}$) as route r^* . For the second child node, route/column r^* is removed from the **RMP**, and the sub-problem corresponding to route r^* is forced to avoid this route.

Specifically, a depth-first strategy is employed to select nodes for branching. This strategy chooses nodes in the reverse order in which they are generated, thereby ensuring that the deepest nodes in the search tree are preferred. The depth-first strategy enables the faster attainment of high-quality feasible solutions to update the node set.

4.5. Acceleration methods

The design of acceleration methods is a critical factor in reducing the computational time. Consequently, several acceleration methods are developed to improve the efficiency of the BPH algorithm.

4.5.1. The improved pulse algorithm for ART sub-problems

The efficiency of solving sub-problems significantly affects the computational time of the BPH algorithm. Directly invoking Gurobi to solve the sub-problem models and obtain the optimal column is a viable approach applicable to the ET sub-problem because it involves fewer constraints and variables. However, the ART sub-problem cannot be solved directly by Gurobi within an acceptable computational time if the number of ART tasks is large. The ART sub-problem can be reduced to a constrained shortest-path problem, which has been proven to be NP-hard and is typically tackled using a label correcting algorithm (Garey and Johnson, 1979; Righini and Salani, 2006). The pulse algorithm, which incorporates a depth-first search and tailored pruning strategies, has been widely applied to enhance the label correcting algorithm (Lozano et al., 2015; Wang and Sheu, 2019). Consequently, we design an improved pulse algorithm (IPA) to efficiently solve the ART sub-problem by employing the following strategies.

- (1) Pruning by infeasibility. The IPA checks if a cycle structure is formed whenever a partial route reaches an unvisited ART task. If such an event occurs, then the partial route visiting the ART task is infeasible and must be pruned.
- (2) Pruning by dominance. Let (i, N_i, c_i) be a label of ART task i in a partial route, where N_i represents the task set in the partial route from beginning task 0 to task i , and c_i represents the reduced cost of the partial route. Given two labels (i, N_i, c_i) and (i, N'_i, c'_i) with the same last visited task i , the latter label is dominated by the former if $N_i \subseteq N'_i$ and $c_i \leq c'_i$. Note that after a partial route visits task i , if the sum of the dual variables for the tasks in the set $C^I \setminus N_i$ does not exceed 0, then this partial route ends because visiting other tasks cannot yield a better reduced cost.
- (3) Greedy extending strategy. In the depth-first search process, the extending strategy determines the acquisition order of partial routes; thus we apply a greedy extending strategy to determine the next visited task for each partial route. The application of this strategy enables the IPA to obtain a high-quality partial route earlier, thereby dominating other non-optimal routes. First, we select the partial route with the lowest reduced cost. We then prioritize extending the dummy termination task and subsequently extend the other unvisited tasks in descending order of dual variable μ_i .
- (4) Operational timing for ART tasks. After a partial route reaches task i , it is necessary to schedule the timing of the empty travel operations for arc $(\hat{i}, i)$ and timing of all operations for task i , where index $\hat{i}$ is the last task of the partial route. In this regard, we formulate a model **SAT_M** for a single ART task, as presented in Appendix B. The timing of all operations in set $O_i \cup O_{ii}^E$ can be obtained by directly using Gurobi to solve the model **SAT_M**. Notably, the IPA does not require invoking Gurobi at each iteration. Based on the dual variable acceleration method described in Section 4.5.3, the operations of some tasks can be performed sequentially without considering resource occupation.

4.5.2. Pre-termination and multi-column addition accelerations

The solution obtained by solving the **RMP** with column generation can provide a lower bound for the AESP. However, the slow convergence of column generation when approaching the optimal solution, known as the long tail effect, can lead to significant computational time (Meng et al., 2015; Xu and Meng, 2020; Liu et al., 2023a). Consequently, we propose a pre-termination heuristic method to prematurely terminate column generation if the gap between the incumbent objective value and the optimal value is within a predefined relative tolerance level σ , $0 \leq \sigma \leq 1$. The following Proposition 2 can thus be obtained.

Proposition 2. (Assume that for an iteration of column generation, the optimal objective value of the **RMP** is D^* , and the corresponding smallest reduced costs of sub-problems **SP_1** and **SP_2** satisfy $P_v^* \geq -\frac{D^* \times \sigma}{L^{SP}}$, $v \in V$ and $P_i^* \geq -\frac{D^* \times \sigma}{L^{SP}}$, $i \in C^E$, respectively, where $L^{SP} = |V| + |C^E|$ is the number of sub-problems. Then $D^* \times (1 - \sigma \times L^{SP})$ is a lower bound for the **MP**.) **Proof.** Define $(\mu_i^*, \pi_v^*, \eta_{hs}^*)_{i \in C^E, v \in V, h \in H, s \in S}$ and $(\chi_{rr'hh's}^*)_{r \in \bar{R}^I, r' \in \bar{R}^E, (h,h') \in \Phi, s \in S}$ as the optimal dual solutions of the **RMP** in the incumbent iteration, where $\bar{R}^I$ and $\bar{R}^E$ represent subsets of sets R^I and R^E for sub-problems **SP_1** and **SP_2**, respectively. Note that $\chi_{hh's}^* = \sum_{r \in \bar{R}^I} \sum_{r' \in \bar{R}^E} \chi_{rr'hh's}^*$. Two additional constraints, $\sum_{r \in \bar{R}^I} \lambda_r \leq L^{SP}$ and $\sum_{r \in \bar{R}^E} \lambda_r \leq L^{SP}$, are introduced to the **RMP**, forming a new model **NRMP**, where λ_r is the decision variable for both **RMP** and **NRMP** and $L^{SP} = |V| + |C^E|$. The additional constraints do not affect the optimality of the solution since the constraints are never activated in the optimal solution, implying that the optimal solution of the **NRMP** is equivalent to that of the **RMP**. Furthermore, we define $P^* = \sum_{v \in V} P_v^* + \sum_{i \in C^E} P_i^*$, where P_v^* and P_i^* indicate the smallest reduced costs for sub-problems **SP_1** and **SP_2**, respectively. Hence, we can obtain the following equations, with r being the optimal route for ART v .

$$P_v^* = w_r - \sum_{i \in C^I} \mu_i^* \alpha_{ir} - \sum_{h \in H} \sum_{s \in S} \eta_{hs}^* \epsilon_{hsr} - \sum_{(h,h') \in \Phi} \sum_{s \in S} \chi_{hh's}^* \epsilon_{hsr} - \pi_v^* \quad (76)$$

$$P_i^* = D_i - \sum_{h \in H} \sum_{s \in S} \eta_{hs}^* \bar{\epsilon}_{hsi} - \sum_{(h,h') \in \Phi} \sum_{s \in S} \chi_{hh's}^* \bar{\epsilon}_{hsi} - \mu_i^* \quad (77)$$

Accordingly, $(P_v^*, P_i^*)_{v \in V, i \in C^E}$, $(\mu_i^*, \pi_v^*, \eta_{hs}^*)_{i \in C^E, v \in V, h \in H, s \in S}$, and $(\chi_{hh's}^*)_{(h,h') \in \Phi, s \in S}$ form a feasible solution for the dual problem of the **NRMP**, and the objective value of this feasible solution is $D^* + L^{SP} \times P^*$. Based on the strong duality theorem, the optimal objective value of the **NRMP** is equal to that of its dual problem, indicating that $D^* + L^{SP} \times P^*$ is a lower bound of the **NRMP** due to the fact that the **NRMP** is a minimization problem. As the **RMP** has the same optimal solution as the **NRMP**, $D^* + L^{SP} \times P^*$ is also a lower bound for the **RMP**. Given $D^* + L^{SP} \times P^* = D^* + L^{SP} \times (\sum_{v \in V} P_v^* + \sum_{i \in C^E} P_i^*) \geq D^* \times (1 - \sigma \times L^{SP})$, $D^* \times (1 - \sigma \times L^{SP})$ is a lower bound of the **RMP** and thus the **MP**. $\square$

Additionally, we introduce a multi-column addition method to further accelerate the algorithm solution process (Liu et al., 2021). In each column generation iteration, we record multiple columns (*Col* columns) with negative reduced costs generated from solving each sub-problem to added to the **RMP**. This acceleration method captures more meaningful columns to reduce the number of iterations, thereby decreasing the computational time of the BPH algorithm.

4.5.3. Other acceleration methods

In this section, we develop two acceleration methods that do not affect the quality of the final solution, including set reduction method and dual variable acceleration method.

First, we describe the set reduction method for the resources and time steps. For different sub-problems, vehicles may occupy different resources, as shown in Fig. 1 and Fig. 4. In the ART sub-problem, the resources include QCs, ARMGs, ARMG handover points for ARTs, QC handover points, and ART paths. In the ET sub-problem, the resources contain ARMGs, ARMG handover points for ETs, and ET paths. In fact, the resources corresponding to different ET tasks stored in different yard blocks are also different, and the resource set of each ET sub-problem can thus be reduced. Moreover, the time step set of each ET sub-problem can be reduced by excluding the time steps before the ready time of the ET. Therefore, the computational burden of each sub-problem, particularly the ET sub-problem, can be alleviated by considering reduced sets of resources and time steps.

The dual variable acceleration method can be explained in two aspects. First, if the sum of all the dual variables corresponding to a sub-problem does not exceed 0, then the sub-problem cannot generate columns with negative reduced costs. As a result, the sub-problem is not required to be solved in the current iteration of the column generation, as presented in rows 6–9 of Algorithm 2. Second, to obtain the optimal operation schemes for individual tasks, we need to repeatedly call Gurobi for directly solving **SAT_M** (of ART tasks) and **SP_2** (of ET tasks). However, when the sum of the dual variables related to resources and time steps (i.e., η_{hs} and $\chi_{hh's}$) for a single task equals 0, the optimal operation scheme is to perform all operations sequentially regardless of resource occupation. Accordingly, we introduce Proposition 3 to demonstrate that the above situation does not affect the optimality of the solutions for **SAT_M** and **SP_2**. This approach, which avoids calling the Gurobi repeatedly, effectively reduces the computational time and accelerates the BPH algorithm.

Proposition 3. (If the sum of the dual variables related to resources and time steps (i.e., η_{hs} and $\chi_{hh's}$) for task i equals 0, then the optimal operation scheme for task i is to perform all operations in sequential order without considering resource occupation.)

Proof. Let the sum of the dual variables be $\Delta = \sum_{h \in H_i} \sum_{s \in S_i} \eta_{hs} + \sum_{(h,h') \in \Phi_i} \sum_{s \in S_i} \chi_{hh's}$, where H_i , S_i , and Φ_i are the resource set, time step set, and resource pair set of task i , respectively. Define $\Phi_i = \{(h, h') | (h, h') \in \Phi, h \in H_i, i \in C^I\}$ and $\Phi_i = \{(h, h') | (h, h') \in \Phi, h' \in H_i\}$,

$i \in C^E$. This proposition can be proven for different types of tasks as follows.

Case 1. If task i is an ART task and $\Delta = 0$, the model **SAT_M** can be transformed into a new model, **N_SAT_M**, as follows. In **N_SAT_M**, all constraints represent the time relationships between different ART operations, and resource occupation has no influence on the objective function. Consequently, the optimal solution of **N_SAT_M** can be obtained by performing all operations of task i sequentially, regardless of the resource occupation.

$$[\mathbf{N_SAT_M}] \quad \text{Min } D_i - \mu_i \quad (78)$$

Subject to constraints (B.2)–(B.9) and (B.19)–(B.20).

Case 2. If task i is an ET task and $\Delta = 0$, model **SP_2** can be transformed into a new model, **N_SP_2**, as follows. Similar to **Case 1**, the objective function is only related to the time after performing all operations of task i , and the optimal solution of **N_SP_2** can thus be generated by sequentially performing all operations of ET task i .

$$[\mathbf{N_SP_2}] \quad \text{Min } D_i - \mu_i \quad (79)$$

Subject to constraints (65)–(69) and (74)–(75). $\square$

5. Computational experiments

In this section, we describe the computational experiments conducted to evaluate the performance of the proposed algorithm. The experiments are implemented on a personal computer with an i7-12700 CPU 2.1 GHz and 16 GB RAM. The solution algorithm is coded using Python, and the commercial software Gurobi 9.5.1 is utilized to solve all linear programming and mixed-integer linear programming models.

5.1. Data description

Based on the physical layout of the ACT in Tianjin Port (China) as shown in Fig. 1, the field survey data, and existing studies (Xu et al., 2021b; Tang et al., 2021; Gao et al., 2022), the relevant data considered for our experiments can be described as follows. The distance from the yard to the terminal apron is 121 m, the width of the ART lane under the QC is 35 m, the landward depth of the yard is 441.5 m, the terminal width is 365 m, the track gauge of a yard block is 34 m, the width of the ART lane between two adjacent yard blocks is 19.5 m, and the width of the ET lane between two adjacent yard blocks is 23 m. On the basis of the analysis in section 3.1, the paths can be categorized into six types based on the length of the paths. According to the data mentioned above and the layout shown in Fig. 4 and Fig. 5, it can be reasonably inferred that the lengths of the six types of paths are 20 m, 34 m, 55 m, 100 m, 5 m, and 10 m, respectively. The speeds of both the ART and ET are set to 5 m/s. Accordingly, the travel time for each vehicle on each path can be deduced. The resource capacity, i.e., the number of vehicles that can be accommodated in each time step, is defined as follows: the capacities of horizontal and vertical paths in the yard are set to 2 and 3, respectively; the capacity of paths under the QC is set to 5; the capacity of vertical paths from the quayside to the yard is set to 3; the capacity of horizontal paths from the quayside to the yard is set to 8; the capacity of paths for ART entering the yard is set to 2; the capacity of paths for ETs entering the yard block is set to 2; the capacities of QCs and ARMGs are both set to 1; the capacity of QC handover points is set to 3; and the capacity of ARMG handover points is set to 2. The QC processing times for ART tasks follow U (60, 120) s, while the ARMG processing times for ART and ET tasks follow U (40, 80) s. The time windows for the ART tasks are provided based on the QC operation list, where the ready and due times of containers belonging to the same QC are sequentially enumerated. The time range for the possible distribution of ET tasks is determined based on the latest time among all the ART task time windows, and the time window of each ET task is then randomly generated within this time range. The yard block corresponding to each container is generated randomly.

The parameters involved in the proposed algorithm are as follows. The relative tolerance level used to pre-terminate the column generation is $\sigma = 0.00001$, and the number of columns added to the **RMP** in each iteration is $Col = 10$. The maximum number of nodes in the BPH algorithm is $B^{node} = 30$, which balances the computational time and solution quality. The computational results for different parameter settings are presented in Appendix C. The duration of each time step is set to 10 s. The solution time limit for the Gurobi is set to 3600 s. In Section 5.2 and 5.3, we demonstrate the effectiveness of the proposed method and perform sensitivity analysis for 1 berth, 4 QCs and 6 yard blocks. To verify the applicability of the BPH algorithm in practice, we conduct computational experiments in Section 5.4 based on the actual layout of the ACT in Tianjin Port, which consists of 3 berths, 12 QCs, 24 yard blocks, and 76 ARTs.

5.2. Effectiveness evaluation of the proposed algorithm

5.2.1. Comparison with Gurobi, TCG, and real port scheduling schemes

To demonstrate the effectiveness of the proposed algorithm, we compare the BPH algorithm with two benchmark methods and two real port scheduling schemes. The first approach involves directly invoking the Gurobi to solve model **M1**. The second approach is a truncated column generation (TCG) algorithm, which has been widely adopted in various domains such as vehicle scheduling, parallel batch scheduling, and liquefied natural gas bunkering station operation problems (Kulkarni et al., 2018; Ozturk, 2020; Liu et al., 2023a). The pseudocode of the TCG is presented in Appendix D. Note that the TCG avoids branching and truncates the column generation process, resulting in a significantly reduced computational time. For a fair comparison, we run the TCG 10 times for each

Table 1

Comparison results for small-scale instances.

Instance	M1(Gurobi)				BPH				TCG			FCFS		EDF	
	LB (s)	Obj (s)	Gap (%)	CPU (s)	LB (s)	Obj (s)	Gap (%)	CPU (s)	Obj (s)	Gap (%)	CPU (s)	Obj (s)	Gap (%)	Obj (s)	Gap (%)
A2_C3_E2	151	151	0.00	107.2	151	151	0.00	6.1	151	0.00	5.1	151	0.00	167	10.60
A2_C4_E2	143	143	0.00	31.3	143	143	0.00	3.6	143	0.00	3.5	183	27.97	159	11.19
A3_C5_E3	166	166	0.00	411.2	166	166	0.00	17.6	166	0.00	14.9	182	9.64	182	9.64
A2_C6_E3	411	411	0.00	1823.0	411	411	0.00	25.8	411	0.00	24.7	447	8.76	448	9.00
A3_C7_E4	204	299	46.57	3600.0	295	299	1.36	56.7	299	1.36	66.8	373	24.75	369	23.41
A4_C7_E5	393	393	0.00	2560.1	389	393	1.03	76.9	393	1.03	62.4	485	23.41	400	1.78
A3_C8_E5	280	443	58.21	3600.0	441	443	0.45	81.2	443	0.45	75.4	572	29.12	511	15.35
A3_C10_E5	207	593	≥100	3600.0	593	593	0.00	82.8	597	0.67	91.4	671	13.15	723	21.92
A4_C10_E6	187	401	≥100	3600.0	393	396	0.76	185.6	409	4.07	166.9	497	25.51	502	26.77
A4_C12_E8	192	921	≥100	3600.0	912	921	0.99	221.4	953	4.50	202.8	1192	29.42	1132	22.91
A5_C12_E8	–	–	–	3600.0	697	705	1.15	256.4	725	4.02	312.9	869	23.26	974	38.16
A5_C15_E10	379	902	≥100	3600.0	857	865	0.93	352.3	879	2.57	313.9	984	13.76	976	12.83
A6_C15_E10	–	–	–	3600.0	706	713	0.99	318.7	741	4.96	270.4	941	31.98	799	12.06
A7_C15_E12	72	834	≥100	3600.0	413	420	1.69	226.4	425	2.91	211.3	581	38.33	485	15.48
A6_C20_E15	–	–	–	3600.0	752	768	2.13	417.6	793	5.45	462.4	1072	39.58	1143	48.83
A7_C20_E15	–	–	–	3600.0	589	589	0.00	394.2	594	0.85	378.1	723	22.75	850	44.31
A6_C25_E18	–	–	–	3600.0	794	811	2.14	494.5	833	4.91	512.7	1019	25.65	1033	27.37
A7_C25_E18	–	–	–	3600.0	746	759	1.74	553.2	762	2.14	564.9	815	7.38	902	18.84
Average	232	471	–	2874.0	525	530	0.85	209.5	540	2.22	207.8	653	21.91	653	20.58

Table 2

Comparison results for large-scale instances.

Instance	BPH				TCG			FCFS		EDF	
	LB(s)	Obj(s)	Gap(%)	CPU(s)	Obj(s)	Gap(%)	CPU(s)	Obj(s)	Gap(%)	Obj(s)	Gap(%)
A5_C30_E30	1155	1174	1.65	596.7	1196	3.55	563.8	1373	16.95	1385	17.97
A6_C30_E30	928	941	1.40	674.3	954	2.80	621.9	1163	23.59	1187	26.14
A7_C30_E30	804	820	1.99	668.2	839	4.35	673.6	971	18.41	932	13.66
A6_C40_E40	1209	1225	1.32	769.2	1256	3.89	748.8	1462	19.35	1527	24.65
A7_C40_E40	1124	1135	0.98	719.1	1153	2.58	716.8	1291	13.74	1206	6.26
A8_C40_E40	976	993	1.74	829.4	1013	3.79	793.4	1276	28.50	1344	35.35
A7_C50_E50	1392	1418	1.87	636.9	1434	3.02	671.6	1698	19.75	1603	13.05
A8_C50_E50	1277	1293	1.25	836.7	1307	2.35	892.8	1507	16.55	1624	25.60
A9_C50_E50	1132	1149	1.50	538.9	1173	3.62	523.4	1675	45.78	1409	22.63
A10_C60_E60	1398	1410	0.86	864.3	1421	1.65	843.8	1780	26.24	1641	16.38
A12_C60_E60	1157	1188	2.68	772.7	1197	3.46	794.4	1362	14.65	1375	15.74
A12_C80_E80	1529	1553	1.57	927.4	1614	5.56	933.8	1651	6.31	1793	15.45
A15_C80_E80	1123	1139	1.42	1019.5	1148	2.23	1107.3	1369	20.19	1467	28.80
A12_C100_E100	1916	1943	1.41	1286.7	1983	3.50	1258.9	2166	11.48	2143	10.29
A15_C100_E100	1725	1759	1.97	1173.0	1764	2.26	1061.3	2037	15.80	2187	24.33
A15_C120_E120	1970	1993	1.17	1467.5	2029	2.99	1394.1	2633	32.11	2641	32.51
A18_C120_E120	1562	1607	2.88	1360.9	1619	3.65	1401.5	2191	36.34	2160	34.41
A18_C150_E150	2497	2561	2.56	1723.8	2593	3.84	1688.7	2906	13.47	3172	23.86
A30_C150_E150	1849	1873	1.30	1634.2	1897	2.60	1793.1	2163	15.48	2208	17.89
A40_C150_E150	1428	1428	0.00	1796.1	1461	2.31	1704.5	1680	17.65	1576	10.36
A50_C180_E180	1637	1693	3.42	2174.9	1693	3.42	2311.3	1806	6.67	1902	12.34
A60_C180_E180	1268	1295	2.13	2038.4	1304	2.84	1839.6	1413	9.11	1570	21.24
A70_C210_E210	1507	1532	1.66	2406.5	1532	1.66	2527.6	1702	11.10	1647	7.51
A80_C210_E210	1162	1187	2.15	2671.2	1203	3.53	2739.8	1484	25.02	1329	11.96
Average	1405	1430	1.70	1232.8	1449	3.14	1233.6	1698	19.34	1710	19.52

instance and record the best solution to achieve a computational time comparable to that of BPH. Furthermore, we introduce two real port scheduling schemes: first-come-first-served (FCFS) and earliest deadline first (EDF). The details of the two schemes are described in [Appendix E](#).

[Table 1](#) presents the comparison results of the proposed algorithm with the Gurobi, TCG, and real port scheduling schemes for small-scale instances. In [Table 1](#), the instance “AX_CY_EZ” represents that the number of ARTs, ART tasks, and ETs for this instance are X, Y, and Z, respectively. For example, Instance “A2_C4_E2” indicates that this instance includes 2 ARTs, 4 ART tasks, and 2 ETs. The symbols “LB”, “Obj”, “Gap”, and “CPU” for the BPH and Gurobi represent the lower bound, objective value, optimality gap ((Obj - LB)/LB × 100 %), and computational time, respectively. Since TCG is a column generation-based heuristic algorithm, it does not provide a better lower bound than that obtained by BPH. Therefore, the “Gap” of the TCG represents the difference between its objective value and the lower bound from the BPH. The symbols “Obj” and “CPU” for the TCG have the same meanings as in BPH. Additionally, the symbols “Obj” and “Gap” for the FCFS and EDF represent the objective value of the scheduling scheme and the optimality gap compared with the BPH result. It can be observed from [Table 1](#) that as the instance size increases, the Gurobi gradually fails to generate solutions and lower bounds within the time limit. Hence, we present only the comparison results of the proposed algorithm with TCG, FCFS, and EDF for large-scale instances in [Table 2](#). Note that to ensure acceptable computational time, we set the time limit of the branch-and-bound process to 10 min in large-scale instances.

The comparison of these methods for small- and large-scale instances can be summarized as follows.

- Directly invoking the Gurobi to solve model **M1** yields an optimal solution for extremely small instances. However, as the problem size increases, Gurobi fails to generate a feasible solution due to reaching the time limit or exhausting the computer memory, and cannot even provide any lower bound information.
- The TCG can obtain high-quality solutions for all instances in a relatively short time, significantly outperforming the Gurobi and two practical scheduling schemes. Furthermore, compared to the lower bounds, TCG achieves average optimality gaps of 2.22 % for small-scale instances and 3.14 % for large-scale instances.
- The BPH algorithm consistently outperform TCG, Gurobi, and two real port scheduling schemes. BPH is particularly effective for medium- and large-scale instances that cannot be solved by Gurobi, and it is obviously better than the two real port scheduling schemes. The average optimality gaps of BPH results compared to FCFS and EDF results under all instances are 20.54 % and 19.87 %, respectively. Additionally, the BPH algorithm outperforms TCG in terms of solution quality for the same computational time cost. Specifically, the BPH algorithm could generate high-quality solutions for all instances within 45 min, with average optimality gaps of 0.85 % and 1.70 % for small- and large-scale instances, respectively. As shown in [Table 2](#), the BPH still outperforms the TCG and two real port scheduling schemes when the number of ARTs reaches 30–80. The outstanding performance of the proposed algorithm indicates that the BPH algorithm is an effective and practical approach that satisfies the requirements of real-world applications.

Moreover, the effects of movement demand on congestion, makespan, and capacity utilization under different solution methods are

Table 3
Performance of the different acceleration methods.

Instance	BPH			BPH_NoIPA			BPH_NoMCA			BPH_NoSR			BPH_NoDVA		
	LB (s)	Obj (s)	CPU (s)	LB (s)	Obj (s)	CPU (s)	LB (s)	Obj (s)	CPU (s)	LB (s)	Obj (s)	CPU (s)	LB (s)	Obj (s)	CPU (s)
A2_C4_E2	143	143	3.6	143	143	11.3	143	143	2.6	143	143	6.2	143	143	10.5
A2_C6_E3	411	411	25.8	411	411	26.5	411	411	27.3	411	411	29.4	411	411	27.8
A3_C7_E4	295	299	56.7	295	299	61.8	295	299	66.5	295	299	71.9	295	299	59.7
A3_C10_E5	593	593	82.8	593	593	116.3	593	593	156.6	593	593	146.3	593	593	125.2
A4_C10_E6	393	396	185.6	393	396	228.6	393	396	207.5	393	396	258.7	393	396	237.9
A5_C12_E8	697	705	256.4	697	705	301.1	697	705	283.4	697	705	310.5	697	705	408.8
A6_C15_E10	706	713	318.7	706	713	546.5	706	713	645.4	706	713	445.2	706	713	716.1
A6_C20_E15	752	768	417.6	752	768	926.6	752	768	701.6	752	768	751.1	752	768	662.4
A7_C25_E18	746	759	553.2	746	759	822.6	746	759	478.8	746	759	789.5	746	759	781.7
A7_C30_E30	804	820	668.2	804	820	1026.8	804	823	827.3	804	820	859.8	804	820	1268.2
A8_C40_E40	976	993	829.4	968	993	1317.9	968	997	756.1	976	993	1092.8	976	993	1194.8
A9_C50_E50	1132	1149	538.9	1132	1149	1198.7	1132	1149	657.3	1132	1149	879.6	1132	1149	1362.5
A10_C60_E60	1398	1410	864.3	1394	1476	2461.8	1394	1425	967.4	1398	1410	1158.7	1398	1410	1609.2
A12_C80_E80	1529	1553	927.4	1529	1580	3600.0	1529	1571	1348.1	1529	1553	1574.9	1529	1553	2445.6
A15_C100_E100	1725	1759	1173.0	1718	1794	3600.0	1718	1802	1728.5	1725	1759	2083.3	1725	1834	3600.0
A18_C120_E120	1562	1607	1360.9	1537	1629	3600.0	1537	1634	1683.0	1562	1607	2462.5	1562	1712	3600.0

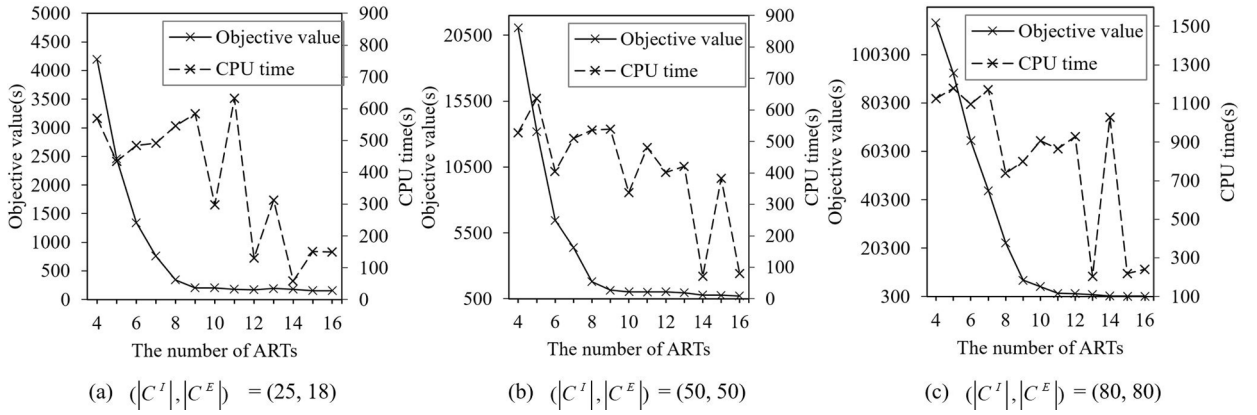


Fig. 9. Effects of the number of ARTs.

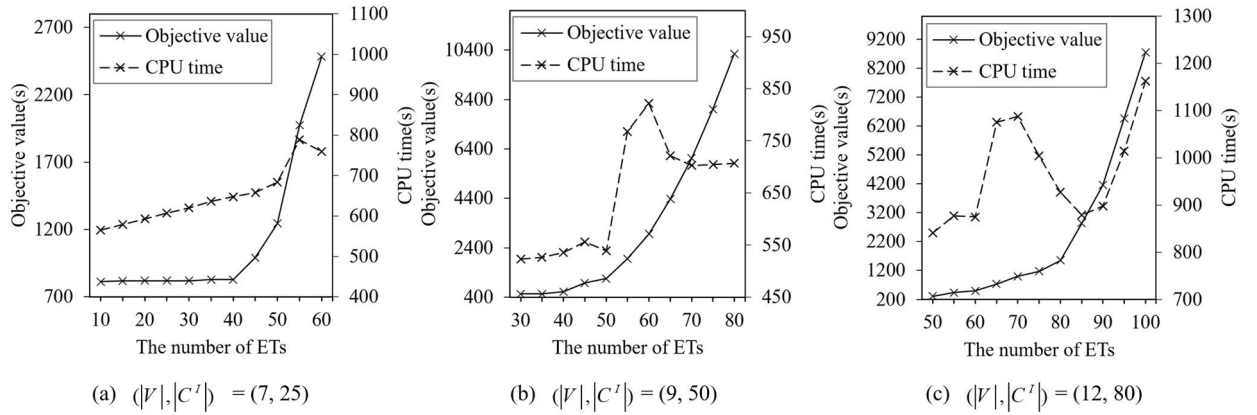


Fig. 10. Effects of the number of ETs.

presented in [Appendix F](#).

5.2.2. Comparison of different acceleration methods

To verify the effectiveness of the proposed acceleration methods, we compare the results of the BPH algorithm equipped with different acceleration methods. The following symbols can thus be specified.

- BPH_NoIPA: BPH without the improved pulse algorithm, where ART sub-problems are solved by calling the Gurobi directly;
- BPH_NoMCA: BPH without the multi-column addition method;
- BPH_NoSR: BPH without the set reduction method;
- BPH_NoDVA: BPH without the dual variable acceleration method.

The comparison results are illustrated in [Table 3](#), where symbols “LB”, “Obj”, and “CPU” indicate the lower bound, objective value, and computational time, respectively. The best computational results are highlighted in bold.

The proposed acceleration methods play a significant role in improving the performance of the BPH algorithm, as listed in [Table 3](#). BPH_NoIPA exhibits a significantly longer computational time compared to BPH, indicating that the improved pulse algorithm effectively speeds up the sub-problem solving process. Compared to adding only one column in each iteration, the multi-column addition method accelerates the update of dual variables and thus reduces the number of iterations. Moreover, this method increases the possibility of generating high-quality columns. Therefore, the performance of BPH_NoMCA is inferior to that of BPH due to the trade-off between column quantity and quality introduced by the multi-column addition method. BPH_NoSR provides objective values comparable to those of BPH; however, the computational time required for BPH is shorter. This is attributed to the fact that the set reduction method reduces the unnecessary search space during the sub-problem solving process. Similarly, the dual variable acceleration method avoids wasting computational resources in unnecessary steps. The performance comparison between the BPH_NoDVA and BPH illustrates that the dual variable acceleration method can effectively enhance the algorithm.

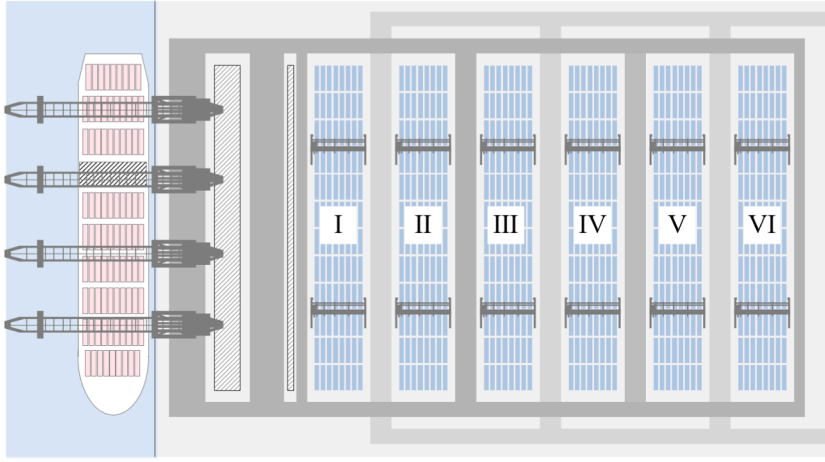


Fig. 11. Diagram of container storage locations.

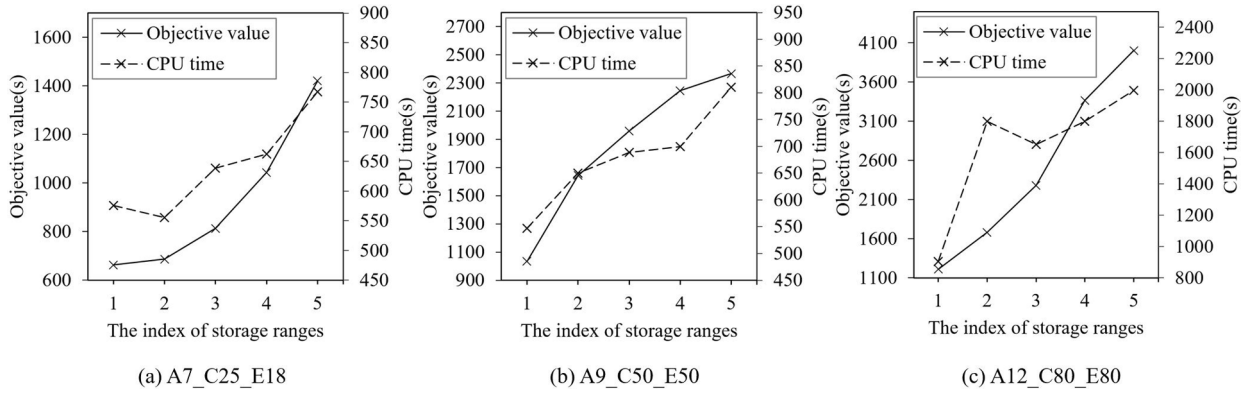


Fig. 12. Effects of storage locations for ART tasks with limited yard resources.

5.3. Sensitivity analysis

5.3.1. Effects of the number of ARTs and ETs

To illustrate the effect of the number of vehicles on the final solution, we explore the effects of the ART and ET quantities on the objective value and computational time. A sensitivity analysis of the number of ARTs is conducted in three instances with ART quantities ranging from 4 to 16, where ART and ET task quantities ($|C^I|$, $|C^E|$) are (25, 18), (50, 50), and (80, 80), respectively. Similarly, a sensitivity analysis of the number of ETs is implemented in three instances with ART and ART task quantities ($|V|$, $|C^I|$) set as (7, 25), (9, 50), and (12, 80), respectively. The ranges of ET quantities in these three instances are 10–60, 30–80, and 50–100. Fig. 9 and Fig. 10 present the effects of the numbers of ARTs and ETs on the performance of the BPH algorithm.

Fig. 9 reveals that with a small number of ARTs, increasing the number of ARTs leads to a significant decrease in the objective value. However, as the number of ARTs further increases, the objective value tends to stabilize at a constant value. This phenomenon occurs because if the number of ARTs is small, delays occur due to the QCs waiting for ARTs. However, if the number of ARTs is sufficient, ARTs wait for QCs, and the limited QC operation capacity becomes the main factor causing delays. An excessive number of ARTs also increases the possibility of vehicle conflicts and resource occupancy conflicts. Moreover, we observe from Fig. 9 that the CPU time remains within 20 min, and the algorithm incurs higher computational costs to when searching for better ART routes when the number of ARTs is small.

The trend observed in Fig. 10 illustrates that an increase in the number of ETs results in an increase in the objective value, with the rate of increase accelerating as the number of ETs increases. This is because ARMGs can satisfy the operational requirements if the number of ETs is small, while an excessive number of ETs leads to a sharp increase in the number of tasks that must be simultaneously performed by limited yard resources, thus creating substantial delays. Additionally, the computational time exhibits an increasing trend with an increase in the number of ETs, indicating that the BPH has a higher computational cost in scheduling yard resources with a larger number of ETs.

Moreover, Appendix F analyzes how variations in the numbers of ARTs and ETs affect congestion, makespan, and capacity

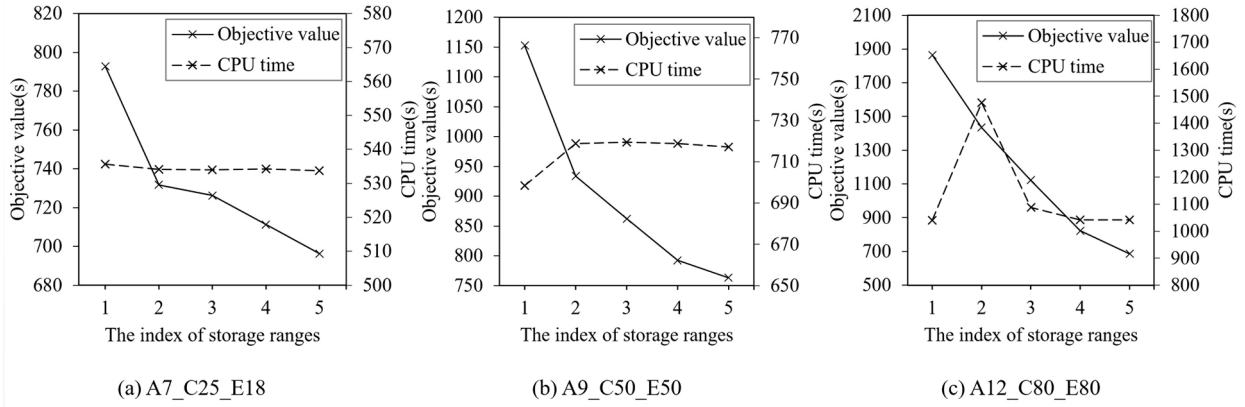


Fig. 13. Effects of storage locations for ET tasks with limited yard resources.

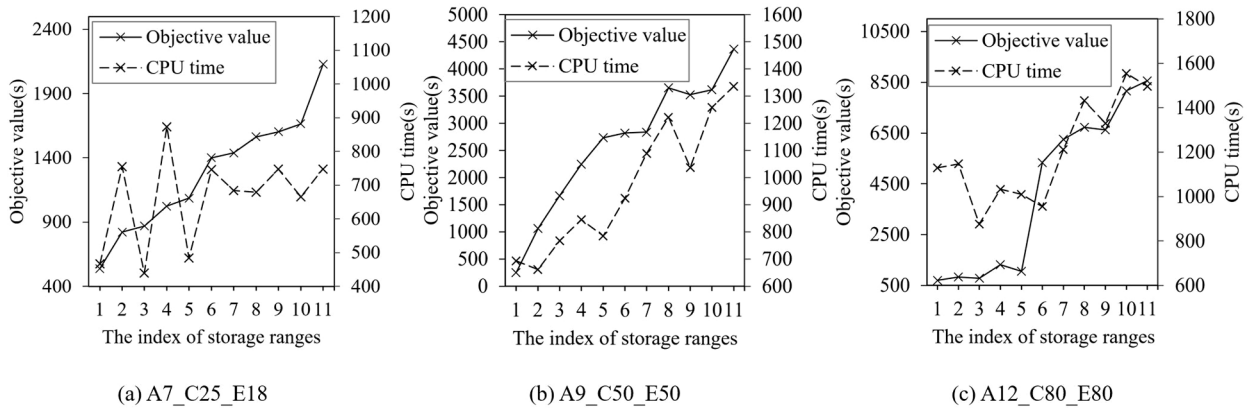


Fig. 14. Effects of storage locations for ART tasks with unlimited yard resources.

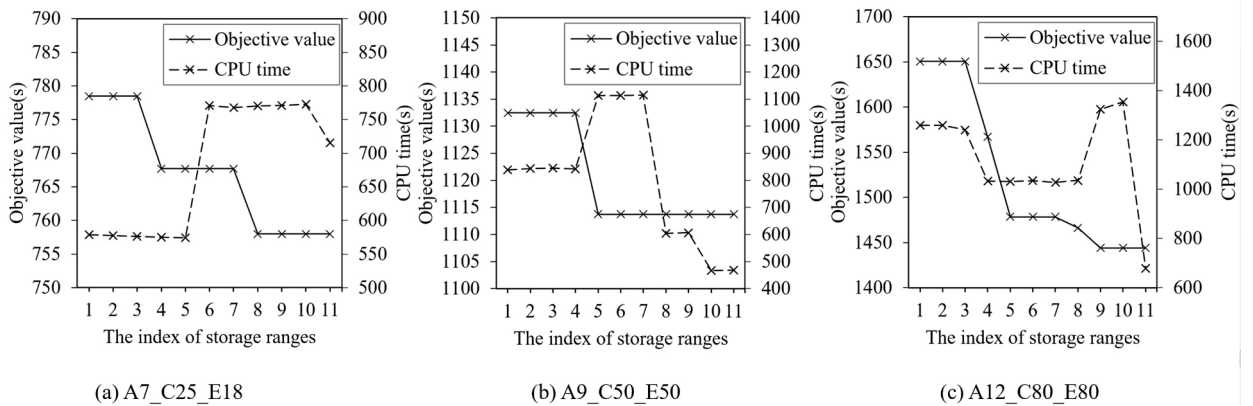


Fig. 15. Effects of storage locations for ET tasks with unlimited yard resources.

utilization.

5.3.2. Effects of the storage locations

The storage locations of the containers in the yard directly affect the travel distance of the vehicles. Moreover, when a large number of containers with similar time windows are stored in the same yard block, the yard resources (ARMGs, HPs, and travel paths) in the block may not be able to meet the operational requirements, resulting in significant delays in container operations. Therefore, we

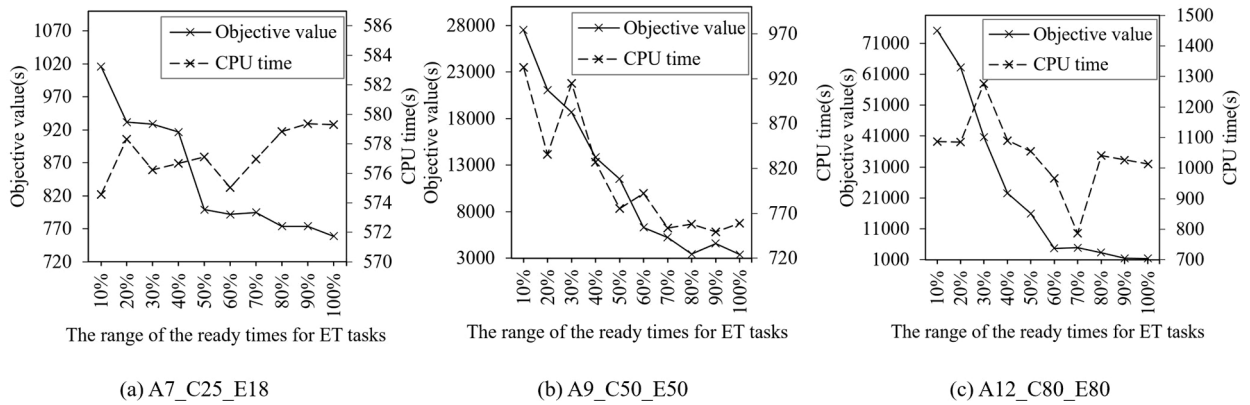


Fig. 16. Effects of the ready times for ET tasks.

investigate the impact of the storage locations of the ART and ET tasks on the performance of the proposed algorithm. As depicted in Fig. 11, containers can be stored in six adjacent yard blocks. Sensitivity analysis of the storage locations is conducted for the ART and ET tasks in three instances: A7_C25_E18, A9_C50_E50, and A12_C80_E80, which can be developed in the following two aspects.

- First, the storage locations of the containers are divided into five ranges: {I, II}, {II, III}, {III, IV}, {IV, V}, and {V, VI}, with each range containing two blocks to ensure equal yard resources. Fig. 12 and Fig. 13 illustrate the results of the objective value and computational time for the ART and ET tasks with limited yard resources, respectively.
- Second, to further analyze the effect of storage locations without considering yard resource limitations, we set the yard resources of all blocks to infinity and divide the storage locations into {I}, {I, II}, {II}, {II, III}, {III}, {III, IV}, {IV}, {IV, V}, {V}, {V, VI}, and {VI}. Fig. 14 and Fig. 15 present the results of the ART and ET tasks with unlimited yard resources, respectively.

The significant effect of the storage location on the total delay time can be seen from Fig. 12 to Fig. 15, and the detailed analysis is presented as follows.

- Storing containers in blocks closer to the quayside implies shorter loaded travel distances for ART tasks but longer total travel distances for ET tasks. Therefore, the objective values in Fig. 12 and Fig. 14 exhibit a significant increase as the storage range index increases, whereas the objective values in Fig. 13 and Fig. 15 decrease as the storage range index increases.
- From Fig. 12 and Fig. 14, it can be observed that regardless of the limited yard resources, the farther the storage locations of the ART tasks from the quayside, the greater the total delay time. This is because ART operations are a continuous process, and waiting for one ART task can result in significant delays for subsequent ART tasks, regardless of whether yard resources are limited.
- By comparing Fig. 13 and Fig. 15, it is evident that the objective value in Fig. 13 varies significantly with changes in the storage locations of the ET tasks, while the objective value in Fig. 15 shows relatively minor variations. This can be attributed to the fact that with limited yard resources, the operations of ET tasks experience delays owing to waiting in the yard, and these delays decrease as the travel distances of the ET tasks decrease. If the yard resources are unlimited, delays in ET tasks occur less frequently and depend solely on the travel distance.
- Furthermore, the BPH algorithm can provide solutions for all instances within an acceptable CPU time. Specifically, storing ART tasks in higher-numbered yard blocks leads to an increase in CPU time, while the storage locations of the ET tasks have a relatively minor impact on the CPU time.

5.3.3. Effects of the ready times for ET tasks

In this section, we conducted further computational experiments to analyze the effects of the ready times for ET tasks on the results. The ready times of the ET tasks are divided into 10 ranges {10 %, 20 %, ..., 100 %}, each of which represents the distribution range of the ready times in the entire planning horizon. For example, the 10 % case indicates that the ready times of the ET tasks is uniformly distributed in the first 10 % of the planning horizon, while the 100 % case indicates that the ready times are uniformly distributed throughout the planning horizon. A sensitivity analysis of the ready times for the ET tasks is conducted for three instances: A7_C25_E18, A9_C50_E50, and A12_C80_E80, and the results are shown in Fig. 16.

Fig. 16 suggests that a more dispersed distribution of ready times leads to smaller objective values. The total delay time is significant when the ready times of the ET tasks are highly concentrated (10 %–60 %), whereas there is little variation in the total delay time when the ready times is dispersed (70 %–100 %). This is because if the ready times of the ET tasks are too concentrated, the limited yard resources cannot handle a large number of tasks simultaneously, thereby leading to substantial delays. Additionally, all instances can be solved within a reasonable computational time.

Table 4

Comparison results of a dynamic instance.

Stage	$(\bar{C}^I , \bar{C}^E)$	BPH		TCG			FCFS		EDF	
		Obj(s)	CPU(s)	Obj(s)	Gap(%)	CPU(s)	Obj(s)	Gap(%)	Obj(s)	Gap(%)
1	(25,25)	121	223.3	125	3.31	215.3	134	10.74	129	6.61
2	(25,25)	254	205.2	269	5.91	219.6	273	7.48	284	11.81
3	(25,25)	136	180.1	141	3.68	171.3	153	12.50	148	8.82
4	(25,25)	274	224.1	290	5.84	181.6	352	28.47	306	11.68
5	(25,25)	162	240.7	167	3.09	162.2	177	9.26	182	12.35
6	(25,25)	74	165.9	78	5.41	241.3	103	39.19	82	10.81
7	(25,25)	29	161.7	29	0.00	204.2	37	27.59	37	27.59
8	(25,25)	176	216.6	181	2.84	195.0	204	15.91	197	11.93
9	(25,25)	73	198.6	73	0.00	202.9	91	24.66	76	4.11
10	(25,25)	147	201.5	152	3.40	236.4	194	31.97	155	5.44
11	(25,25)	108	221.2	113	4.63	209.4	113	4.63 %	127	17.59
12	(25,25)	68	235.2	72	5.88	174.5	94	38.24	79	16.18
13	(25,25)	174	170.1	178	2.30	158.9	204	17.24	180	3.45
14	(25,25)	130	199.5	134	3.08	233.7	138	6.15	166	27.69
15	(25,25)	293	174.5	304	3.75	214.3	314	7.17	322	9.90
16	(25,25)	131	219.6	137	4.58	178.6	161	22.90	149	13.74
17	(25,25)	218	200.4	227	4.13	179.8	239	9.63	262	20.18
18	(25,25)	74	181.7	76	2.70	213.4	92	24.32	81	9.46
19	(25,25)	62	189.6	65	4.84	204.8	68	9.68	71	14.52
20	(25,25)	116	190.3	121	4.31	179.5	131	12.93	125	7.76
Average	–	141	200.0	147	3.68	198.8	164	18.03	158	12.58

Table 5

Comparison results for dynamic real-world scale instances.

Instance	$(\bar{C}^I , \bar{C}^E)$	BPH		TCG			FCFS		EDF	
		Obj(s)	CPU(s)	Obj(s)	Gap(%)	CPU(s)	Obj(s)	Gap(%)	Obj(s)	Gap(%)
C600_E600	(20,20)	89	64.3	91	2.25	63.2	98	10.11	103	15.73
C600_E600	(15,15)	96	39.8	99	3.13	37.6	104	8.33	112	16.67
C600_E600	(10,10)	105	23.2	108	2.86	24.1	115	9.52	127	20.95
C600_E600	(5,5)	109	9.4	113	3.67	8.9	121	11.01	124	13.76
C1200_E1200	(20,20)	126	57.8	130	3.17	58.3	146	15.87	148	17.46
C1200_E1200	(15,15)	135	40.2	136	0.74	38.5	152	12.59	145	7.41
C1200_E1200	(10,10)	133	22.9	144	8.27	21.8	149	12.03	163	22.56
C1200_E1200	(5,5)	137	8.6	146	6.57	9.2	158	15.33	180	31.39
C3000_E3000	(20,20)	176	60.2	183	3.98	61.7	214	21.59	196	11.36
C3000_E3000	(15,15)	180	39.6	191	6.11	38.4	226	25.56	214	18.89
C3000_E3000	(10,10)	193	23.5	198	2.59	22.9	206	6.74	227	17.62
C3000_E3000	(5,5)	203	9.3	215	5.91	9.0	243	19.70	229	12.81
C6000_E6000	(20,20)	235	59.7	250	6.38	60.7	289	22.98	263	11.91
C6000_E6000	(15,15)	247	38.4	258	4.45	37.8	264	6.88	279	12.96
C6000_E6000	(10,10)	253	22.5	268	5.93	21.2	282	11.46	276	9.09
C6000_E6000	(5,5)	278	8.3	290	4.32	8.7	314	12.95	325	16.91
Average	–	–	–	–	4.40	–	–	13.92	–	16.09

5.4. Case study

The practicality of the proposed approach is verified using a real-world ACT, namely, the Beijiang Section C automated container terminal located in Tianjin Port, China. The terminal has 3 berths, 12 QCs, 24 yard blocks, 42 ARMGs, and 76 ARTs, as depicted in Fig. 1. The total quayside length is 1100 m, and the yard area covers 750,000 m². The remaining detailed data related to the terminal layout and data generation rules related to the container parameters are provided in Section 5.1. In practice, the ACT commonly handles thousands of containers, where the equipment conditions and operational environment are uncertain and dynamically changing. Consequently, providing a one-time operation plan for thousands of containers is unnecessary because it cannot adapt to dynamic environmental changes and is challenging to accomplish within a reasonable computational time. Accordingly, based on the actual case of the ACT at Tianjin Port, we conduct computational experiments on real-world scale instances in a dynamic environment.

In a dynamic instance, the container information is provided in stages rather than all at once. Therefore, we divide the dynamic instance into multiple stages and utilize the proposed algorithm to obtain a solution for each stage. At the end of each stage, the locations and times of ART, as well as the resource occupancy information, are recorded as input parameters for the subsequent stage. To illustrate how real-time results are obtained for the dynamic instance, we provide a dynamic instance with 500 ART and 500 ET tasks, as shown in Table 4.

In Table 4, we schedule 25 ART and 25 ET tasks at each stage, resulting in a total of 20 stages for the dynamic instance. The results

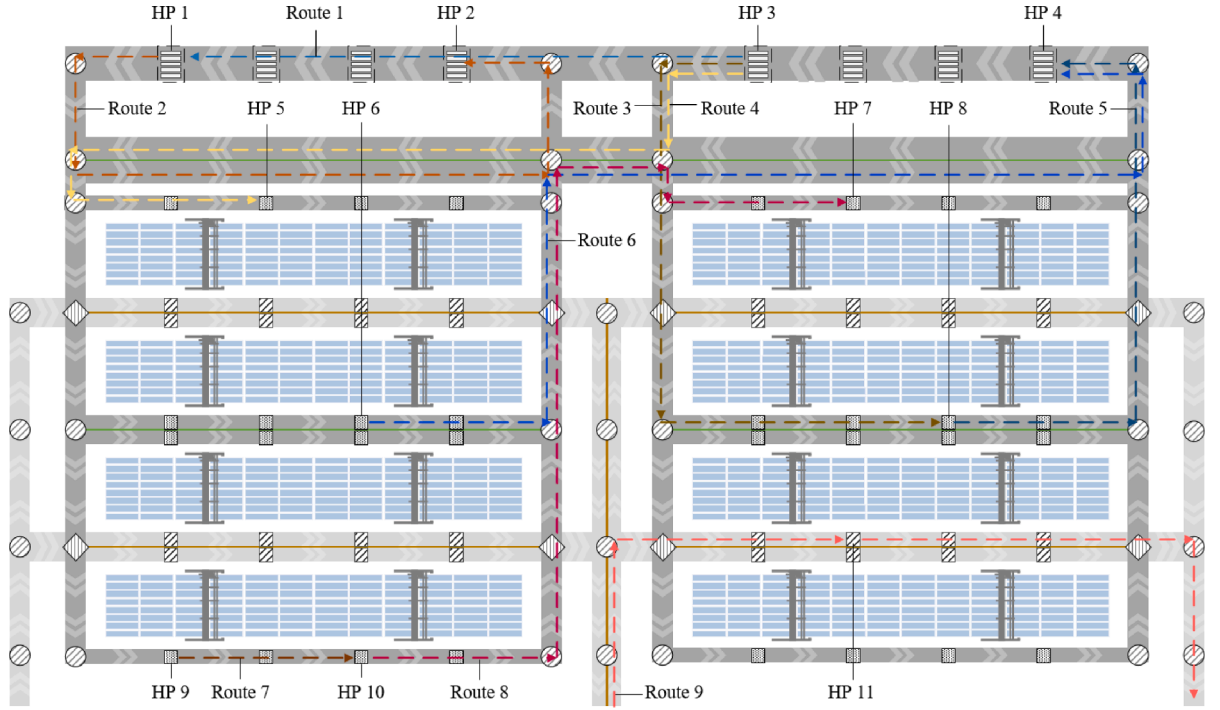


Fig. A1. The examples of ART and ET travel routes.

Table C1

Sensitivity analysis for the different values of parameter Col .

Instance	BPH($Col = 5$)		BPH($Col = 10$)		BPH($Col = 15$)		BPH($Col = 20$)		BPH($Col = 25$)	
	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)
A2_C4_E2	143	3.1	143	3.6	143	3.2	143	3.7	143	3.9
A3_C5_E3	166	16.4	166	17.6	166	17.3	166	17.1	166	17.4
A3_C7_E4	299	55.4	299	56.7	299	56.6	299	57.1	299	57.3
A4_C10_E6	396	158.2	396	185.6	396	226.9	396	196.8	396	183.1
A5_C12_E8	705	190.1	705	256.4	705	229.6	705	285.3	705	234.6
A6_C15_E10	713	343.6	713	318.7	713	352.1	713	383.6	713	398.4
A7_C20_E15	589	364.4	589	394.2	589	445.8	589	476.2	589	513.9
A8_C40_E40	997	793.4	993	829.4	993	973.2	993	1087.7	993	1304.5
A9_C50_E50	1149	485.1	1149	538.9	1149	614.8	1149	694.8	1149	803.1
A10_C60_E60	1425	991.2	1410	864.3	1410	927.6	1418	1236.9	1418	1407.2
A15_C80_E80	1139	1218.1	1139	1019.5	1168	1173.9	1168	1396.2	1168	1419.8
A18_C150_E150	2584	1437.9	2561	1723.8	2561	2078.6	2603	1815.4	2603	2174.8
A30_C150_E150	1903	1371.8	1873	1634.2	1873	1809.4	1873	2125.7	1873	2397.4
A50_C180_E180	1718	1918.6	1693	2174.9	1684	2276.9	1684	1961.8	1684	2536.3
A60_C180_E180	1327	2391.7	1295	2038.4	1295	2463.2	1295	2269.7	1295	2817.5
A80_C210_E210	1235	2706.4	1187	2671.2	1187	3017.8	1174	3159.1	1174	3406.2

for each stage obtained using TCG, FCFS, and EDF are presented in Table 4. Sets $\bar{C}^f$ and $\bar{C}^E$ represent the newly obtained ART task set and ET task set at each stage, respectively. The symbols “Obj” and “CPU” indicate the objective value and computational time for each approach in the corresponding stage. The symbol “Gap” for each comparison approach is the optimality gap between the objective value of this approach and that of BPH algorithm. As can be seen from Table 4, the proposed BPH algorithm can generate scheduling results for a dynamic instance within reasonable computational time. The results obtained by the BPH are significantly better than those of the TCG and the two real port scheduling schemes, with average optimality gaps of 3.68 %, 18.03 %, and 12.58 %.

To further validate the applicability of the proposed algorithm to different dynamic real-world scale instances, we solve dynamic instances with ART and ET task quantities of (600, 600), (1200, 1200), (3000, 3000), and (6000, 6000). The number of ART and ET tasks ($|\bar{C}^f|$, $|\bar{C}^E|$) for each stage is set to (20, 20), (15, 15), (10, 10), and (5, 5). Accordingly, the results for the 16 instances are presented in Table 5, where “Obj” and “CPU” of each approach represent the average objective value and computational time for all stages, respectively. The results in Table 5 demonstrate that a larger number of ART and ET tasks considered in each stage implies a higher solution quality and longer computational time. The BPH algorithm proves to be an effective approach for dynamically solving real-

Table C2Sensitivity analysis for the different values of parameter B^{node} .

Instance	BPH($B^{node} = 10$)		BPH($B^{node} = 20$)		BPH($B^{node} = 30$)		BPH($B^{node} = 40$)		BPH($B^{node} = 50$)	
	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)
A2_C4_E2	143	3.6	143	3.2	143	3.6	143	3.5	143	3.1
A3_C5_E3	166	17.4	166	17.4	166	17.6	166	17.5	166	17.3
A3_C7_E4	299	56.2	299	56.1	299	56.7	299	56.9	299	56.1
A4_C10_E6	401	138.4	396	183.6	396	185.6	396	189.7	396	188.5
A5_C12_E8	711	169.3	711	207.4	705	256.4	705	328.3	701	391.4
A6_C15_E10	737	206.1	713	249.0	713	318.7	713	384.1	713	453.9
A7_C20_E15	589	390.9	589	395.7	589	394.2	589	398.2	589	396.1
A8_C40_E40	1018	508.4	1011	674.9	993	829.4	993	1063.4	993	1271.5
A9_C50_E50	1173	417.2	1149	473.6	1149	538.9	1138	691.8	1138	816.5
A10_C60_E60	1421	591.3	1421	683.1	1410	864.3	1410	1268.4	1410	1339.4
A15_C80_E80	1148	718.9	1139	891.4	1139	1019.5	1139	1369.2	1139	1576.1
A18_C150_E150	2607	1107.9	2593	1397.3	2561	1723.8	2543	2193.7	2529	2419.5
A30_C150_E150	1909	1006.7	1897	1287.1	1873	1634.2	1873	1937.4	1873	2218.3
A50_C180_E180	1716	1538.1	1716	1869.4	1693	2174.9	1693	2579.6	1693	2993.7
A60_C180_E180	1304	1373.2	1304	1674.2	1295	2038.4	1283	2317.5	1279	3008.2
A80_C210_E210	1218	1893.4	1187	2346.7	1187	2671.2	1187	3267.1	1187	3528.6

Table C3Sensitivity analysis for the different values of parameter σ .

Instance	BPH($\sigma=0.0005$)		BPH($\sigma=0.0001$)		BPH($\sigma=0.00005$)		BPH($\sigma=0.00001$)		BPH($\sigma=0.000005$)	
	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)	Obj(s)	CPU(s)
A2_C4_E2	143	3.7	143	3.1	143	3.5	143	3.6	143	3.2
A3_C5_E3	166	17.7	166	18.2	166	18.7	166	17.6	166	18.1
A3_C7_E4	299	56.4	299	56.1	299	57.3	299	56.7	299	58.2
A4_C10_E6	396	158.4	396	160.7	396	161.9	396	185.6	396	190.4
A5_C12_E8	705	225.9	705	221.7	705	254.8	705	256.4	705	293.4
A6_C15_E10	713	267.7	713	272.1	713	308.9	713	318.7	713	376.2
A7_C20_E15	589	301.3	589	348.7	589	389.1	589	394.2	589	401.6
A8_C40_E40	1011	780.7	1011	789.6	1011	812.3	993	829.4	993	1087.7
A9_C50_E50	1149	464.3	1149	481.1	1149	519.6	1149	538.9	1149	739.5
A10_C60_E60	1421	724.2	1421	740.8	1421	746.9	1410	864.3	1410	1248.4
A15_C80_E80	1146	722.6	1146	718.7	1139	936.5	1139	1019.5	1131	1498.1
A18_C150_E150	2603	1293.8	2581	1584.6	2581	1604.3	2561	1723.8	2561	1719.6
A30_C150_E150	1897	1402.7	1897	1433.1	1873	1618.7	1873	1634.2	1873	1697.8
A50_C180_E180	1693	1931.1	1693	2037.6	1693	2116.9	1693	2174.9	1693	2279.5
A60_C180_E180	1304	1643.9	1295	1837.6	1295	2033.7	1295	2038.4	1286	3013.6
A80_C210_E210	1218	2263.1	1218	2519.3	1218	2496.8	1187	2671.2	1187	2946.9

Table F1

Comparison results of congestion and makespan.

Instance	BPH		TCG		FCFS		EDF	
	Congestion (time)	Makespan (s)	Congestion (time)	Makespan (s)	Congestion (time)	Makespan (s)	Congestion (time)	Makespan (s)
A2_C4_E2	9	1013	9	1013	9	1049	9	1076
A3_C5_E3	6	809	6	809	5	817	5	817
A3_C7_E4	4	1092	4	1092	3	1193	4	1104
A4_C10_E6	13	1104	12	1112	13	1125	15	1251
A5_C12_E8	14	1132	13	1180	10	1219	7	1224
A6_C15_E10	34	1143	30	1186	28	1201	24	1269
A7_C20_E15	46	1664	43	1702	34	1766	43	1725
A8_C40_E40	73	2813	67	2907	58	3209	64	3223
A9_C50_E50	100	3621	89	3724	62	3927	61	3966
A10_C60_E60	113	4892	104	4986	78	5027	93	5129
A15_C80_E80	169	5329	162	5391	153	5536	144	5576
A18_C150_E150	273	8726	259	8904	248	9314	229	9658
A30_C150_E150	364	8482	349	8607	328	9031	307	9207
A50_C180_E180	507	9846	465	10,207	443	10,625	458	10,739
A60_C180_E180	596	9721	571	10,095	538	10,446	549	10,671
A80_C210_E210	804	11,192	762	11,597	729	12,679	670	13,041

Table F2
Comparison results of capacity utilization for different resources.

Instance	BPH				TCG				FCFS				EDF			
	QC (%)	ARMG (%)	HP (%)	Path (%)	QC (%)	ARMG (%)	HP (%)	Path (%)	QC (%)	ARMG (%)	HP (%)	Path (%)	QC (%)	ARMG (%)	HP (%)	Path (%)
A2_C4_E2	10.91	10.86	2.63	2.51	10.91	10.86	2.63	2.51	10.36	10.36	1.98	2.24	10.07	10.42	1.78	2.17
A3_C5_E3	16.27	21.36	3.94	3.42	16.27	21.36	3.94	3.42	15.74	18.06	4.38	3.52	15.45	17.73	4.21	3.41
A3_C7_E4	15.92	18.39	3.66	3.78	15.92	18.39	3.66	3.78	13.75	16.56	3.06	3.19	14.86	17.57	3.30	3.45
A4_C10_E6	21.04	26.92	6.01	5.13	20.79	26.68	5.93	4.98	20.33	26.00	5.62	4.85	18.45	22.62	4.75	4.57
A5_C12_E8	27.39	29.61	9.03	5.97	26.09	28.94	8.96	5.81	25.91	28.66	7.20	5.61	26.52	28.35	8.87	5.62
A6_C15_E10	31.16	35.08	13.25	7.22	30.90	34.52	13.09	7.18	30.63	33.44	12.77	7.09	28.24	31.47	11.21	6.74
A7_C20_E15	31.07	35.80	16.41	7.24	30.83	35.57	16.38	7.21	29.66	34.75	16.33	6.91	30.22	35.22	16.07	7.09
A8_C40_E40	33.29	37.93	15.67	7.82	32.96	37.68	15.14	7.69	32.44	37.21	14.33	7.19	32.79	37.09	13.16	7.31
A9_C50_E50	40.57	45.62	21.33	8.38	37.26	43.09	19.46	8.04	32.63	40.46	17.67	7.62	32.92	40.55	18.36	7.55
A10_C60_E60	32.28	41.27	21.26	8.14	31.04	40.53	20.70	7.82	30.26	39.11	19.78	7.57	30.12	39.18	19.19	7.70
A15_C80_E80	38.49	50.06	44.03	10.13	37.04	48.92	43.21	9.87	35.41	47.50	42.77	9.47	35.28	47.24	42.80	9.32
A18_C150_E150	42.09	49.36	38.07	11.28	40.53	46.09	37.16	10.53	37.46	44.93	36.44	9.24	36.47	43.82	35.80	8.81
A30_C150_E150	45.94	53.28	43.94	13.06	43.86	51.93	41.62	12.75	41.66	49.03	39.89	12.17	42.04	49.76	39.97	12.44
A50_C180_E180	47.33	54.79	46.13	14.11	45.87	52.91	45.05	13.84	44.25	51.69	43.61	13.04	44.53	51.26	42.89	12.77
A60_C180_E180	49.06	55.81	49.02	14.94	48.15	54.09	47.48	14.35	45.11	53.18	46.87	14.12	44.83	53.02	46.48	13.96
A80_C210_E210	48.92	55.26	55.63	16.89	46.72	53.61	54.62	16.43	44.27	51.84	53.49	16.02	44.09	50.74	52.84	15.85

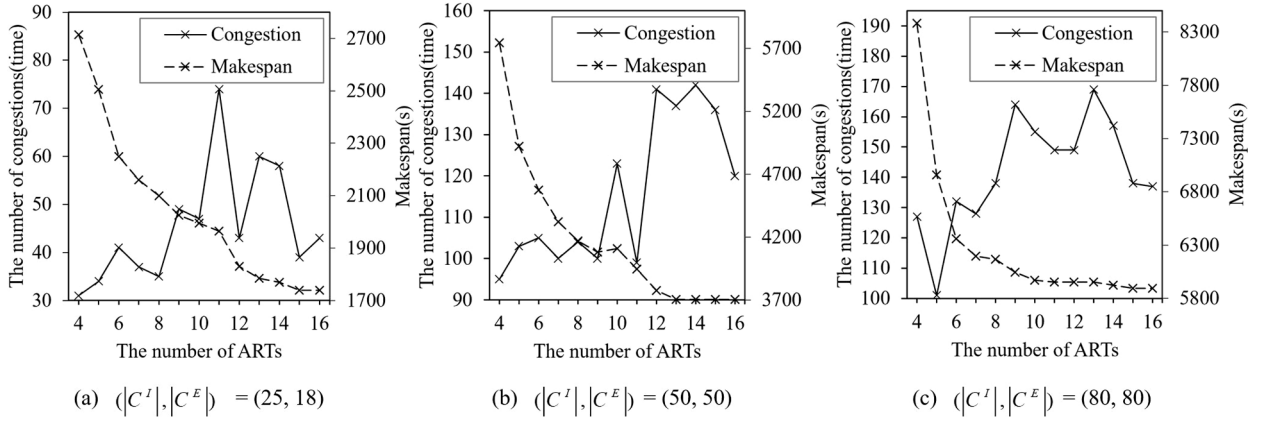


Fig. F1. Effects of the number of ARTs on congestion and makespan.

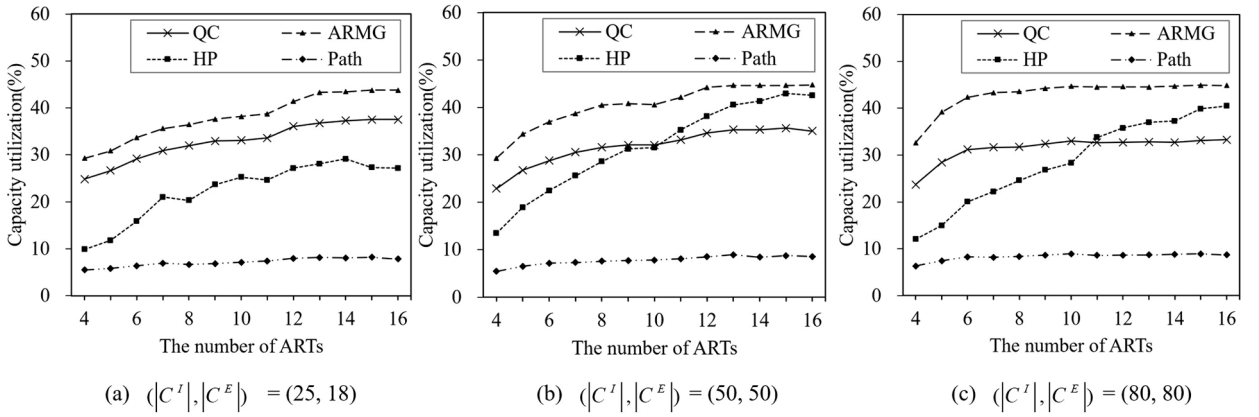


Fig. F2. Effects of the number of ARTs on capacity utilization.

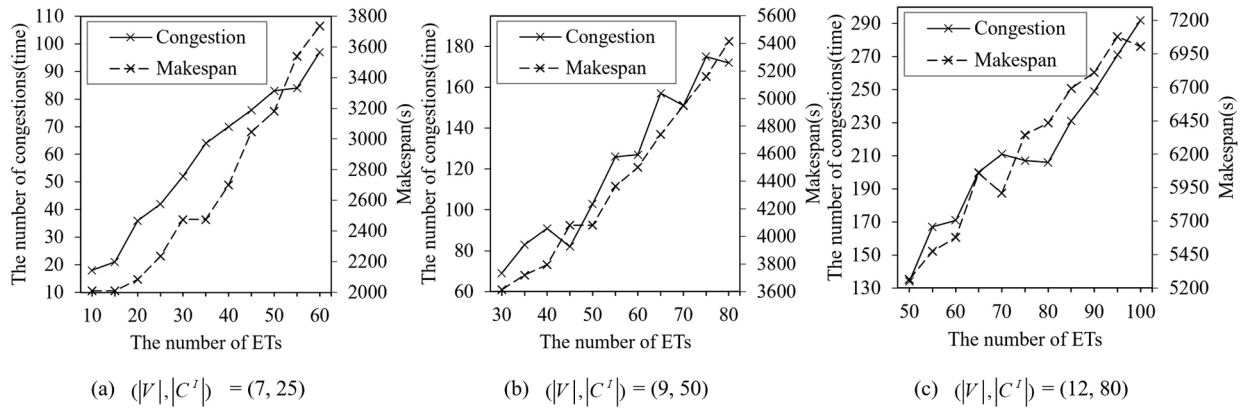


Fig. F3. Effects of the number of ETs on congestion and makespan.

world instances and efficiently addressing ART and ET scheduling problems in the ACT of Tianjin Port. Even for the instance with 6000 ART tasks and 6000 ET tasks, the BPH algorithm can obtain results for each stage within 1 min. Furthermore, the proposed BPH algorithm outperforms the TCG and two real port scheduling schemes, with average optimality gaps of 4.40 %, 13.92 %, and 16.09 %, respectively.

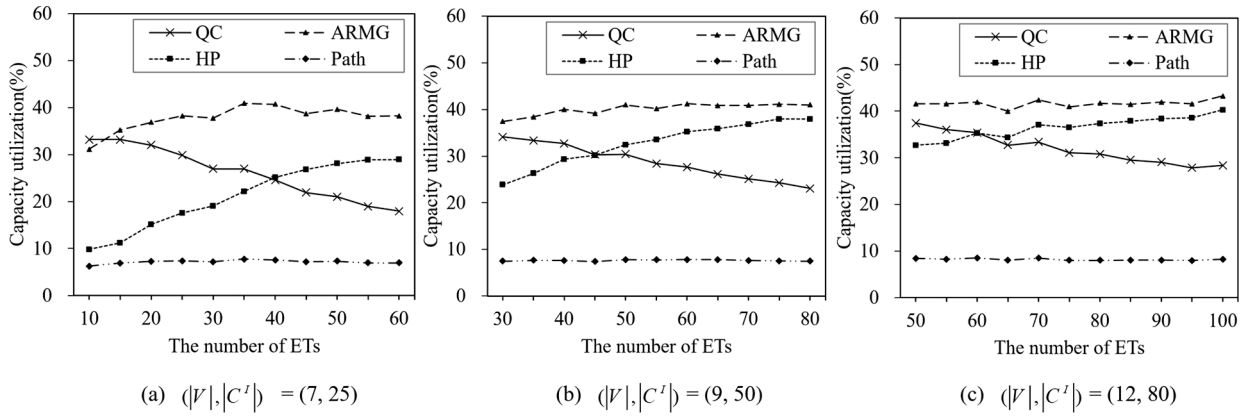


Fig. F4. Effects of the number of ETs on capacity utilization.

6. Managerial insights

This study investigates the ART and ET scheduling problem in a parallel-layout ACT, where lanes exist on both sides of the yard block. To solve this problem, a MILP model is constructed and a tailored BPH algorithm is developed. The effectiveness and applicability of the proposed approach are validated through computational experiments. Based on the background of the studied problem and sensitivity analysis, we provide some managerial insights for port operators.

First, ports should select an appropriate layout and vehicle organization for ACTs based on their specific needs. The perpendicular layout is a classic layout form, while the parallel layout has gained widespread adoption in recent years. 1) In the classic perpendicular layout, the transportation areas for internal trucks and ETs are located at opposite yard ends, creating a complete physical separation between internal trucks and ETs. 2) Considering the high movement requirements of yard cranes in the classic perpendicular layout, a U-shaped perpendicular layout has attracted attention. In the U-shaped layout, internal and external trucks can enter the side of the yard block, allowing the yard crane to focus on container pickup and placement without moving to the ends of the yard. Note that internal trucks and ETs also travel on completely different lanes in this layout. 3) In the parallel-layout ACT with lanes on both sides of the yard block, internal trucks and ETs travel on one side of the block. However, although internal trucks and ETs travel on different horizontal lanes, inevitable intersections occur in vertical lanes. 4) Conventional container terminals typically adopt the parallel layout with lanes on only one side of the yard block. In this layout, interactions between internal trucks and ETs occur on all lanes. In fact, such conventional terminals can undergo automation modifications for cranes and vehicles, so as to transition to semi-ACTs or even fully ACTs. In summary, port operators should consider their actual circumstances and requirements when deciding on the layout and vehicle organization for ACTs.

Second, port operators should consider the impact of the number of vehicles, storage locations, and arrival times of ETs on the operational performance. Based on the sensitivity analysis in Section 5.3 and Appendix F, we provide managerial insights for the studied parallel-layout ACT. 1) The selection of an appropriate number of ARTs is crucial for the ACT. The computational experiments in Section 5.3.1 are conducted with 1 berth and 4 QCs, as described in Section 5.1. The results from Fig. 9 and Fig. F1 indicate that both the objective value curve and makespan curve stabilize when the number of ARTs reaches 9–12. From Fig. F2, it can be observed that an increase in the number of ARTs cannot increase the crane utilization indefinitely, but causes vehicle waiting in the HP. Consequently, a ratio of 1:3 to 1:4 between the number of QCs and ARTs is desirable to ensure that the number of ARTs is adequate to meet the operational requirements without being excessive. 2) It is crucial to maintain a reasonable range of the number of ETs arriving within a given planning horizon, to ensure full utilization of yard resources without causing significant delays in tasks. An excessive number of ETs can exacerbate the situation of vehicles waiting in the HP. 3) To improve the overall operational efficiency of the terminal, it is advisable to store ART tasks in yard blocks closer to the quayside, whereas ET tasks should be stored in blocks farther away from the quayside. In addition, the storage locations of the ET tasks should be dispersed rather than concentrated in the same block to mitigate delays caused by limited yard resources. 4) In practice, the ready times of ET tasks can be considered as the arrival times of ETs. As a result, in the actual operation of an ACT, the arrival times of ETs should be as scattered as possible, and the number of arriving ETs in each time period should preferably match the yard operation capacity.

7. Conclusions

In this study, we investigate the ART and ET scheduling problem in an ACT with a parallel layout. Considering vehicle congestion, limited HP capacity, limited crane operation capacity, and the interaction between the two types of vehicles, the problem determines the task assignment of the ARTs and the timing of all operations for the ARTs and ETs. To address this problem, we propose an MILP model that involves a significant number of big-M constraints and polynomial decision variables. Further, by analyzing the problem characteristics, we adapt the original model into a route-based model that utilizes exponential variables without big-M constraints. To address the computational challenges in solving the model, we develop a tailored BPH algorithm and suggest several acceleration

methods. The experimental results of the BPH algorithm are compared with those of Gurobi, TCG, and two real port scheduling schemes, and the comparison results indicate that the proposed algorithm significantly outperforms the benchmark methods. Moreover, the effectiveness of the acceleration methods in enhancing algorithm performance is validated. Specifically, we conduct a case study of a real-world ACT, which illustrates the ability of our proposed approach to dynamically solve the problem based on new information in a dynamically changing environment. Based on the problem background and computational experiments, some valuable managerial insights are provided.

The current study can be extended in several directions. First, this study assumes that the speeds of ARTs and ETs are known; however, in practice, their speeds may vary during starting, stopping, and turning. Therefore, an interesting research direction is to consider the effects of acceleration and deceleration on travel time, and to centrally control the travel speeds of vehicles (especially ARTs). Second, since the storage locations of containers have a significant influence on the ART and ET scheduling results, future research could incorporate storage location decisions to further extend our model. Third, this study assumes that all input parameters, including the container exchange times between vehicles and cranes and the ready times of the ETs, are known in advance. However, these parameters may fluctuate because of inclement weather or unexpected events. Therefore, future research can explore robust optimization or stochastic programming methods to consider these uncertain parameters. Alternatively, online optimization methods can be developed to satisfy the rescheduling requirements of the ACTs.

CRedit authorship contribution statement

Zehao Wang: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Qingcheng Zeng:** Conceptualization, Funding acquisition, Investigation, Supervision, Writing – original draft, Writing – review & editing. **Xingchun Li:** Conceptualization, Data curation, Investigation, Writing – review & editing. **Chenrui Qu:** Conceptualization, Funding acquisition, Investigation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partly supported by: (1) the National Natural Science Foundation of China under Grant 72101043; (2) the National Key Research and Development Program of China under Grant 2023YFE0113200; and (3) the Natural Science Foundation of Liaoning Province of China under Grant 2023-BS-072.

Appendix A. The proof of proposition 1

Proof. For ART tasks, the combinations of the start HP and end HP can be categorized into the following four types, each of which is exemplified in Fig. A1.

Type 1. From QC handover point to QC handover point. There are two scenarios for this type, represented by Route 1 and Route 2 in Fig. A1. Route 1 goes from HP 3 to HP 1, while Route 2 goes from HP 1 to HP 2.

Type 2. From QC handover point to ARMG handover point. There are two scenarios for this type, represented by Route 3 and Route 4 in Fig. A1. Route 3 goes from HP 3 to HP 8, while Route 4 goes from HP 3 to HP 5.

Type 3. From ARMG handover point to QC handover point. There are two scenarios for this type, represented by Route 5 and Route 6 in Fig. A1. Route 5 goes from HP 8 to HP 4, while Route 6 goes from HP 6 to HP 4.

Type 4. From ARMG handover point to ARMG handover point. There are two scenarios for this type, represented by Route 7 and Route 8 in Fig. A1. Route 7 goes from HP 9 to HP 10, while Route 8 goes from HP 10 to HP 7.

For ET tasks, there is only one scenario. As shown in Fig. A1, for an ET task with an ARMG handover point at HP 11, the route can be identified as Route 9. It can be observed that since the lanes under cranes are only accessible to vehicles exchanging containers with these cranes, Route 1 to Route 9 in Fig. A1 depict all the possible vehicle travel paths. Therefore, the travel paths of an ET are fixed once the HP for the ET to perform its yard operation is determined, and the ART paths between any two HPs are actually fixed and known. $\square$

Appendix B. The model for a single ART task

Define parameter ξ_i as the completion time of task i by the partial route. Let H_i denote the set of resources occupied by extending task i . Let S_i denote the set of time steps generated based on current time ξ_i . Define a resource pair set $\Phi_i = \{(h, h') | (h, h') \in \Phi, h \in H_i\}$. With these new parameter and sets, the model SAT_M for task i can be formulated as follows.

$$[\text{SAT_M}] \quad \text{Min } D_i - \sum_{h \in H_i} \sum_{s \in S_i} \eta_{hs} \bar{e}_{hsii} - \sum_{(h, h') \in \Phi_i} \sum_{s \in S_i} \chi_{hh's} \bar{e}_{hsii} - \mu_i \quad (\text{B.1})$$

Subject to

$$T_i^F \leq T_{ip_i^0}^L - t_{p_i^0} \quad (\text{B.2})$$

$$T_{ip_i^T}^L \leq T_i^S - g_i^S \quad (\text{B.3})$$

$$T_{iip_i^T}^E \leq T_i^F - g_i^F \quad (\text{B.4})$$

$$\xi_i \leq T_{iip_i^0}^E - t_{p_i^0} \quad (\text{B.5})$$

$$a_i \leq T_i^F - g_i^F \quad (\text{B.6})$$

$$T_i^S - b_i \leq D_i \quad (\text{B.7})$$

$$T_{iip}^E \leq T_{iip}^E - t_p, \forall (p, p') \in P_{ii}^A \quad (\text{B.8})$$

$$T_{ip}^L \leq T_{ip}^L - t_p, \forall (p, p') \in P_i^A \quad (\text{B.9})$$

$$Y_{os} + Z_{os}^B + Z_{os}^A = 1, \forall o \in O_i \cup O_{ii}^E, \forall s \in S_i \quad (\text{B.10})$$

$$\bar{e}_{hsii} = \sum_{o \in (O_i \cup O_{ii}^E) \cap O_h} Y_{os}, \forall h \in H_i, \forall s \in S_i \quad (\text{B.11})$$

$$U_o^B = \begin{cases} T_{ip}^L - t_p, & p = h_o, \forall o \in O_i^L \\ T_{iip}^E - t_p, & p = h_o, \forall o \in O_{ii}^E \\ T_i^F - g_i^F, & o = o_i^F \\ T_i^S - g_i^S, & o = o_i^S \\ T_{iip_i^T}^E, & o = o_i^A \\ T_{ip_i^T}^L, & o = o_i^B \end{cases} \quad (\text{B.12})$$

$$U_o^F = \begin{cases} T_{ip}^L, & p = h_o, \forall o \in O_i^L \\ T_{iip}^E, & p = h_o, \forall o \in O_{ii}^E \\ T_i^F, & o = o_i^F \\ T_i^S, & o \in \{o_i^S, o_i^B\} \\ T_{ip_i^0}^L - t_{p_i^0}, & o = o_i^A \end{cases} \quad (\text{B.13})$$

$$U_o^B \leq e_{s+1} + M(1 - Y_{os}), \forall o \in O_i \cup O_{ii}^E, \forall s \in S_i \quad (\text{B.14})$$

$$e_s \leq U_o^F + M(1 - Y_{os}), \forall o \in O_i \cup O_{ii}^E, \forall s \in S_i \quad (\text{B.15})$$

$$U_o^F \leq e_s + M(1 - Z_{os}^B), \forall o \in O_i \cup O_{ii}^E, \forall s \in S_i \quad (\text{B.16})$$

$$e_{s+1} \leq U_o^B + M(1 - Z_{os}^A), \forall o \in O_i \cup O_{ii}^E, \forall s \in S_i \quad (\text{B.17})$$

$$\bar{e}_{hsii}, Y_{os}, Z_{os}^B, Z_{os}^A \in \{0, 1\}, \forall h \in H_i, \forall s \in S_i, \forall o \in O_i \cup O_{ii}^E \quad (\text{B.18})$$

$$T_{ip}^L, D_i, T_i^F, T_i^S, U_o^B, U_o^F \geq 0, \forall p \in P_i, \forall o \in O_i \cup O_{ii}^E \quad (\text{B.19})$$

$$T_{iip}^E \geq 0, \forall p \in P_{ii} \quad (\text{B.20})$$

The objective function (B.1) is to minimize the reduced cost of the partial route from task i' to task i , similar to equation (43). Constraints (B.2)–(B.13) replicate constraints (48)–(51) and (53)–(60), but only for the operations of arc (i', i) and task i . Constraints

(B.14)–(B.17) are copies of constraints (21)–(24), where operation set O and time step set S are reduced to $O_i \cup O_{it}^E$ and S_i , respectively. Constraints (B.18)–(B.20) specify the domains of decision variables. Note that in this model, the completion time of operation o_i^B is the completion time of the second crane operation for task i , and the time steps in which operation o_i^B occupies resource $h_{o_i^B}$ will be updated in the subsequent extension of the partial route.

Appendix C. Sensitivity analysis of parameter settings

We select 16 test instances from small- and large-scale instances to conduct the sensitivity analysis of parameter settings. Table C1, Table C2, and Table C3 illustrate the effect of changes in parameters Col , B^{node} , and σ on the results, respectively. Parameters Col , B^{node} , and σ vary in $\{5, 10, 15, 20, 25\}$, $\{10, 20, 30, 40, 50\}$, and $\{0.0005, 0.0001, 0.00005, 0.00001, 0.000005\}$, respectively. In Table C1 to Table C3, symbols “LB”, “Obj”, and “CPU” represent the lower bound, objective value, and computational time of the BPH algorithm for different parameter values. The best computational results are highlighted in bold.

In Table C1, the results with a Col value of 10 are superior to those with a Col value of 5, and the difference in computational time between the two is insignificant. The objective value shows similar performance as the Col value changes from 10 to 25. In particular, the BPH algorithm consistently obtains the best solutions in most instances when parameter Col is set to 10. However, as the Col value increases, the computational time increases significantly. Consequently, we choose $Col = 10$ as the best parameter setting.

Table C2 indicates that with the increase in parameter B^{node} , the objective value gradually decreases, while the computational time significantly increases. The solution quality notably improves when parameter B^{node} increases from 20 to 30. As B^{node} continues to increase from 30 to 50, the results are similar in most instances, but the computational time increases significantly. To balance solution quality and computation time, we choose $B^{node} = 30$ as the best parameter setting.

Table C3 shows that as the value of parameter σ increases, the objective value decreases while the computational time increases. When σ is 0.00001, the proposed algorithm can obtain high-quality solutions within an acceptable time, achieving a balance between solution quality and computational time. Therefore, we choose $\sigma = 0.00001$ as the best parameter setting.

Appendix D. The truncated column generation

Truncated column generation (TCG) is a column generation-based heuristic algorithm, where the algorithm is truncated in advance if the termination criteria in Section 4.5.2 is met. The procedure of the TCG is introduced in Algorithm D1. Note that if the decision variable values of the columns in rows 8 and 10 are fixed to 1, the corresponding sub-problems are removed.

Appendix E. Two real port scheduling schemes

Two real port scheduling schemes are described as follows.

- (1) First-come-first-served scheme, FCFS. In this scheme, the vehicle that reaches the idle state earliest is given priority to perform the task with the earliest ready time. For an ART, the idle state is reached after completing its previous task, while the moment for an ET to reach the idle state is the ready time of its corresponding ET task. If an ART is selected, then the task with the earliest ready time among all unassigned ART tasks is detected, and the ART sequentially performs the empty travel, first crane operation, loaded travel, and second crane operation of the selected task. If the selected vehicle is an ET, the ET sequentially performs the travel before yard operation, yard operation, and travel after yard operation of the corresponding ET task. Note that in case of resource occupancy conflicts or vehicle conflicts, subsequent task operations must wait for the operations of the assigned tasks.
- (2) Earliest deadline first scheme, EDF. This scheme begins by selecting the task with the earliest due time. If the selected task is an ET task, its operations are performed sequentially. If an ART task is chosen, then the ART that reaches the idle state earliest is selected to conduct the task, and the operations of the task are performed sequentially. It should be pointed out that similar to the FCFS, the operations of the tasks scheduled first have priority in occupying the corresponding resource.

Appendix F. Congestion, makespan, and capacity utilization results

In addition to comparing the efficiency of solution methods, we analyze the effects of movement demand on congestion, makespan, and capacity utilization from two perspectives. First, we compare the effects of movement demand under different solution methods. Second, the effects of variations in the numbers of ARTs and ETs are analyzed. The three metrics are calculated as follows. 1) Congestion refers to the number of times vehicles on a path reach its capacity. Specifically, for each time step and each path, if the number of vehicles on the path equals its capacity in the time step, the total congestion count increases by one. 2) Makespan represents the time when the last task has been completed. 3) Capacity utilization is calculated for four types of resources: QCs, ARMGs, HPs, and paths. The capacity utilization of resource h in time step s is $\sum_{o \in O_h} Y_{os} / d_h$, where $\sum_{o \in O_h} Y_{os}$ is the number of vehicles for resource h in time step s and d_h is the capacity of resource h . Thus, the average utilization ω_h of resource h per time step can be obtained by Equation (F.1), where G is the set of time steps from the start to the completion time, $S' \subseteq S$. For the BPH and TCH algorithms, the capacity utilization ω_h can also be calculated by Equation (F.2). Note that the values of variables Y_{os} and λ_r are known in the final solution. The final capacity utilization of QCs can be obtained by averaging the utilization of all QC resources. Similarly, the final capacity

utilizations of ARMGs, HPs, and paths can be obtained by averaging the utilization of all resources of the same type.

$$\omega_h = \sum_{s \in S} \sum_{o \in O_h} Y_{os} / (d_h \cdot |S'|) \quad (\text{F.1})$$

$$\omega_h = \sum_{s \in S} \sum_{r \in R} \varepsilon_{hsr} \lambda_r / (d_h \cdot |S'|) \quad (\text{F.2})$$

To compare the effects of movement demand on congestion, makespan, and capacity utilization under different solution methods, we select 16 test cases from the small- and large-scale instances, as shown in [Table F1](#) and [Table F2](#). Note that when the number of ART tasks exceeds 10, the Gurobi method typically fails to generate results within the time limit ([Table 1](#)). However, congestion and capacity utilization are often more valuable for larger-scale scenarios. Therefore, the results of the Gurobi method are not provided in the tables. The comparisons of [Table F1](#) and [Table F2](#) can be summarized as follows.

- In terms of makespan, the solutions obtained by BPH and TCG are significantly better than those obtained by FCFS and EDF. The BPH and TCG methods perform similarly in terms of makespan, with BPH being superior to TCG.
- In terms of congestion, the number of congestion times increases gradually with the size of the instance. In addition, methods with lower makespan correspond to higher congestion times. The reason is that solutions with lower makespan operate the same burden of tasks in a shorter time, resulting in a higher number of congestion times in the path.
- The capacity utilization of resources shows an increasing trend as the instance size increases. Specifically, the capacity utilization ranking for the four methods is as follows: BPH > TCG > FCFS > EDF. Moreover, it can be observed that none of the resources are utilized more than 60 %. The reason is that some resources (especially cranes) have completed their assigned tasks in the later stages of the operational process, but the whole operational process may not conclude because other resources still have unfinished tasks. In an example with two QCs (QC 1 and QC 2), if tasks of QC 1 are completed but QC 2 still has unfinished tasks, QC 1 remains idle until QC 2 finishes its operation.

We then analyze the effects of varying the number of ARTs and ETs on congestion, makespan, and capacity utilization. The instance settings and the range of quantity variations for AETs and ETs are the same as described in [Section 5.3.1](#). [Fig. F1](#) and [Fig. F2](#) illustrate

Algorithm C1 Truncated column generation

Input: data file

Output: final feasible solution Ξ^{TCG} and its objective value D^{TCG}

01: obtain initial columns based on the algorithm presented in [Section 4.3.3](#)

02: **do**

03: conduct rows 3–21 of [Algorithm 2](#) to obtain δ

04: **if** meet the termination criteria described in [Section 4.5.2](#) **or** $\delta = 0$ **then**

05: **if** generate an integer solution **then**

06: $\delta^{TCG} = 0$

07: **else if** exist columns with decision variable values equal to 1 **then**

08: add these columns to the final feasible solution Ξ^{TCG} and fix these variables to 1

09: **else**

10: add the column with the largest variable value to Ξ^{TCG} and fix its variable to 1

11: **end if**

12: **if** the RMP is infeasible **then**

13: remove the columns just added to Ξ^{TCG} and fix their variables to 0

14: **end if**

15: **end if**

16: **loop until** $\delta^{TCG} = 0$

17: **return** Ξ^{TCG} and D^{TCG}

the effects of ART quantity changes on congestion, makespan, and capacity utilization. Similarly, Fig. F3 and Fig. F4 demonstrate the effects of ET quantity changes. The comparisons of Fig. F1 to Fig. F4 can be summarized as follows.

- Fig. F1 indicates that an increase in the number of ARTs significantly reduces the makespan when the number of ARTs is small. However, as the number of ARTs continues to increase, the makespan stabilizes. This trend is similar to the delay time in Fig. 9. In addition, the number of congestion times shows a fluctuating upward trend with an increase in the number of ARTs, which implies that the increase in the number of ARTs exacerbates congestion.
- In Fig. F2, the capacity utilization of resources continues to rise as the number of ARTs increases. Specifically, the growth rate of the crane utilization gradually slows down, while the growth rate of the HP utilization is faster than that of the crane utilization. This is because the operational capacity of cranes is limited, and the sustained increase in the number of ARTs intensifies the waiting of ARTs in the HP, which leads to a significant increase in HP utilization. Additionally, the capacity utilization of paths shows a relatively small range of increase, attributable to the large number of paths in the terminal.
- Fig. F3 shows that with the increase in the number of ETs, both congestion times and makespan continue to rise. A higher number of ETs imposes a greater operational burden on the yard, and the limited yard operation capacity results in significant delays. In addition, the congestion level significantly increases with the growth in the number of ETs.
- In Fig. F4, the changes in the capacity utilization of the four resources vary with the increase in the number of ETs. 1) The QC utilization gradually decreases with the growth in the number of ETs. This is because the increase in the number of ETs leads to an increase in makespan, while ET tasks do not involve QC operations. Therefore, when the QC operation time remains unchanged and makespan increases, the QC utilization decreases. 2) The ARMG utilization increases with the increase in ETs, but the growth rate gradually slows down. The reason is that the increase in the number of ETs adds to the operational burden for ARMGs, while the operational capacity of ARMGs is limited. 3) The HP utilization continues to increase with the increase in ETs. This is attributed to the fact that the limited ARMG capacity makes it more likely for vehicles to wait in the HP as the number of ETs increases. 4) With the increase in the number of ETs, the path utilization remains relatively constant. This is because the increase in the number of ETs leads not only to an increase in the number of vehicles in paths, but also to an increase in makespan. The simultaneous growth in both keeps the path utilization changing little.

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